MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COUNTER, BINARY, SYNCHRONOUS, DIGITAL, KEY CODE FDA

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic synchronous binary counter module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.095 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance and requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FDA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/6-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

* 3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

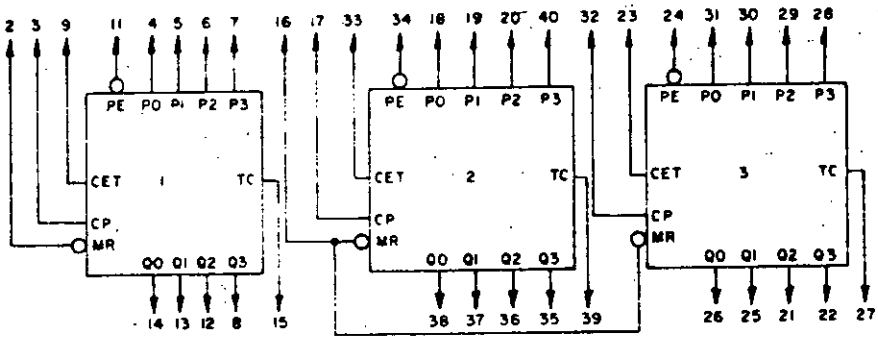
4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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1 ← V_{CC} = 5 V ± 10%

10 ← GND = 0 V REF

NOTES:
1. P0, P1, P2, P3 = Preset inputs.
2. PE = Preset enable.
3. CP = Clock input.
4. CET = Count enable.
5. TC = Terminal count.
6. MR = Master reset.

FIGURE 1. FDA module function diagram, synchronous binary counter.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/ -3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A15	V_{OUT} (low loaded)	III	S	10	800	10	800	10	800	10	800	mV
B1	Input breakdown current	IV	S	0	2	0	2	0	2	0	2	mA
B2-B23	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	mA
C1-C23	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D	Isolation	VI	S	10	∞	1	∞	10	∞	1	∞	MΩ
E1-E3	Functional, high level	VII	X	←			See table VII		→			
F1,F3,F5	Functional, low level	VII	X	←			See table VII		→			
F2,F4,F6	Functional, low level	VII	X	0	400	0	400	0	400	0	400	mV
G1-G15	Functional and static output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
H1-H12	Functional and static output voltage, low level	VII	X	0	400	0	400	0	400	0	400	mV
J1-J3	Input threshold check	VII	X	←			See table VII		→			
K1-K3	Input threshold check	VII	S	←			See table VII		→			
L1	Input current, low level	VIII	X	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	-0.1	-1.73	mA
L2-L11	Input current, low level	VIII	X	-0.1	-3.46	-0.1	-3.46	-0.1	-3.46	-0.1	-3.46	mA
L12-L23	Input current, low level	VIII	X	-0.1	-1.15	-0.1	-1.15	-0.1	-1.15	-0.1	-1.15	mA
M1	Input current, high level	VIII	X	0	40	0	40	0	40	0	40	µA
M2-M11	Input current, high level	VIII	X	0	80	0	80	0	80	0	80	µA

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M12-M23	Input current, high level	VIII	X	0	30	0	30	0	30	0	30	µA
N1-N15	Output short circuit current	IX	S	-20	-80	-20	-80	-20	-80	-20	-80	mA
P1	Power supply current, high level	X	S	50	290	50	290	50	290	50	290	mA
P2	Power supply current, low level	X	S	50	290	50	290	50	290	50	290	mA
Q1	Input capacitance	XI	Q	2	30	2	30	2	30	2	30	pF
Q2-Q23	Input capacitance	XI	Q	2	15	2	15	2	15	2	15	pF
R	Power supply transient amplitude (5 V)	XII	Q	0	400	0	400	0	400	0	400	mV
S1-S6	Transition time (t_{THL})	XIII	S	2	20	2	20	2	25	2	25	ns
S7-S15	Transition time (t_{THL})	XIII	Q	2	20	2	20	2	25	2	25	ns
T1-T6	Transition time (t_{TLH})	XIII	S	4	35	4	35	4	40	4	40	ns
T7-T15	Transition time (t_{TLH})	XIII	Q	4	35	4	35	4	40	4	40	ns
U1,U3,U5	Propagation delay time (t_{PHL})	XIII	S	10	20	10	20	10	22	10	22	ns
U2,U4, U6-U15	Propagation delay time (t_{PHL})	XIII	S	2	45	2	45	2	50	2	50	ns
V1-V3	Propagation delay time (t_{PLH})	XIII	S	4	70	4	70	4	75	4	75	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
V4-V6,V8, V9,V11,V12, V14,V15	Propagation delay time (t_{PLH})	XIII	S	4	50	4	50	4	55	4	55	ns
V7,V10,V13	Propagation delay time (t_{PLH})	XIII	S	12	28	12	28	12	30	12	30	ns
W1-W3	Propagation delay time (t_{PHL})	XIV	S	5	60	5	60	5	70	5	70	ns
W4-W15	Propagation delay time (t_{PHL})	XIV	Q	5	60	5	60	5	70	5	70	ns
X1-X3	Propagation delay time (t_{PHL})	XV	S	2	35	2	35	2	45	2	45	ns
Y1-Y3	Propagation delay time (t_{PLH})	XV	S	2	40	2	40	2	50	2	50	ns
Z1-Z6	Maximum counting frequency	XVI	S	460	500	460	500	460	500	460	500	ns
AA1-AA3	Setup time (CET) check	XVII	S	15.6	16.4	15.6	16.4	15.6	16.4	15.6	16.4	µs
BB1-BB3	Hold time (CET) check	XVII	S	0	400	0	400	0	400	0	400	mV
CC1-CC3	Setup time (P) check	XVIII	S	0.8	1.2	0.8	1.2	0.8	1.2	0.8	1.2	µs
CC4-CC12	Setup time (P) check	XVIII	Q	0.8	1.2	0.8	1.2	0.8	1.2	0.8	1.2	µs
DD1-DD3	Hold time (P) check	XVIII	S	0	400	0	400	0	400	0	400	mV
DD4-DD12	Hold time (P) check	XVIII	Q	0	400	0	400	0	400	0	400	mV
EE1-EE3	Setup time (PE) check	XVIII	S	0.8	1.2	0.8	1.2	0.8	1.2	0.8	1.2	µs
FF1-FF3	Hold time (PE) check	XVIII	S	0	400	0	400	0	400	0	400	mV
FF4-FF12	Hold time (PE) check	XVIII	Q	0	400	0	400	0	400	0	400	mV
GG1-GG15	Output crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	mV

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
HH1-HH15	Output crosstalk, low level	XIX	Q	0.0	1.2	0.0	1.2	0.0	1.2	0.0	1.2	V
HH16-HH30	Output crosstalk, low level	XIX	Q	0	400	0	400	0	400	0	400	mV
J.1	Power supply transient rise time	XX	S	2.5	7.5	2.5	7.5	2.5	7.5	2.5	7.5	µs

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through JJ shall be performed in any order with the following exceptions. If test conditions A, B, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, D where applicable, followed by the remaining tests in the sequence. If test conditions A and B are successively repeated, the time interval between performance of these test conditions shall be a minimum of 10 seconds to preclude detrimental effects due to module heating.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through JJ in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C, E through P, and S through JJ shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Transient thermal test. The transient thermal test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition JJ, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Mechanical shock, thermal shock, humidity, salt fog, and durability tests. Following the environmental tests, the electrical tests of table I which are marked with an (X) in the test type column, and test condition JJ, shall be performed using the initial limits at 25° ±5°C ambient.

* 4.5.4 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated at 60°C (+5°, -0°C) as shown on figure 2. Following the life test, electrical tests C through JJ shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

* 6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.1.1 Setup time. The minimum time a signal must be present on the input before the clock low-to-high transition occurs for operation.

6.1.2 Hold time. The minimum time a logic level must be maintained following the clock low-to-high transition in order to insure recognition.

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6.1.3 Recovery time. Recovery time is defined as the minimum time required between the reset low-to-high transition and the clock low-to-high transition in order to recognize and transfer a high data level to the Q output.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military Specification MIL-M-28787/6 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 2658826. Part number M28787/6-1 supersedes part numbers 2658826 and M28787/6. These parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Supply voltage		5 V dc ±10%
Maximum power dissipation		
All output at low level		1,600 mW (outputs not loaded)
All output at high level		1,600 mW (outputs not loaded)
Clock input		
Frequency		16.6 MHz maximum
Pulse width		30 ns minimum
Input threshold voltage		
Low level		0.80 V maximum
High level		2.0 V minimum
Input current, low level		
MR 1	(V_{IL} = 0.0 V)	-1.73 mA maximum
	(V_{IL} = 0.4 V)	-1.6 mA maximum
CP, PE, CET, MR2, MR3	(V_{IL} = 0.0 V)	-3.46 mA maximum
	(V_{IL} = 0.4 V)	-3.2 mA maximum
P0, P1, P2, P3	(V_{IL} = 0.0 V)	-1.15 mA maximum
	(V_{IL} = 0.4 V)	-1.07 mA maximum
Input current, high level		
MR 1	(V_{IH} = 2.4 V)	40 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
CP, PE, CET	(V_{IH} = 2.4 V)	80 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
P0, P1, P2, P3	(V_{IH} = 2.4 V)	30 µA maximum
	(V_{IH} = 5.5 V)	1.0 mA maximum
MR2, MR3	(V_{IH} = 2.4 V)	80 µA maximum
	(V_{IH} = 5.5 V)	2.0 mA maximum
Output voltage		
Low level (I_{OL} = 10 mA)		0.4 V maximum
High level (I_{OH} = -0.36 mA)		2.4 V minimum
Output drive current maximum capability		
Low level (V_{OL} = 0.4 V)		10 mA
High level (V_{OH} = 2.4 V)		-0.36 mA
Master reset pulse width		30 ns minimum
Master reset recovery time		30 ns minimum

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6.4 Module description - continued.

Propagation delay time, t_{PHL} (C_L = 100 pF)	
CP to Q0-Q3	2 ns minimum, 50 ns maximum
MR to Q0-Q3 or TC	5 ns minimum, 70 ns maximum
CET to TC	2 ns minimum, 45 ns maximum
CP to TC	10 ns minimum, 22 ns maximum
Propagation delay time, t_{PLH} (C_L = 100 pF)	
CP to Q1-Q3	4 ns minimum, 55 ns maximum
CP to Q0	12 ns minimum, 30 ns maximum
CET to TC	2 ns minimum, 50 ns maximum
CP to TC	4 ns minimum, 75 ns maximum
Setup time	
CET	25 ns minimum
P0, P1, P2, P3	40 ns minimum
PE	45 ns minimum
Hold time	
CET	0 ns maximum
P0, P1, P2, P3	-5 ns maximum
PE	-7 ns maximum

* 6.5 Module function. This module contains three 4-bit synchronous binary counters with independent inputs and outputs with the exception that circuits 2 and 3 have a common master reset (MR) input.

6.5.1 Input to output relationship. Each counter has the input to output relationship of table II for positive logic.

6.5.1.1 Master reset input. The asynchronous master reset (MR) input sets the counter output to a low level independent of any other input.

6.5.1.2 Counter. The counter is synchronously presettable when the preset enable (PE) input is low. Data can enter the preset inputs (P0-P3) when the clock input (CP) is low. During the low-to-high transition of CP, the data inputs are inhibited so that a latter change in the input data will not affect the outputs, then the now stored information is transferred to the outputs. During the high-to-low transition of CP, the transfer path from preset inputs to the outputs are inhibited leaving the outputs steady in their preset state, then the preset inputs are enabled so that new data can enter. Some restrictions are placed on the manner of selection. First the transition of the count enable (CET) input from high-to-low or of PE from low-to-high may only be done when CP is high. The remaining transitions may be made following the setup and hold times specified.

6.5.1.3 TC (terminal count) = CET·Q0·Q1·Q2·Q3. The TC output is asynchronous, therefore, the TC output should never be used outside the counting circuit as a synchronous output unless resynchronizing circuitry is added to compensate for the delay differences between the TC and Q outputs. When connected as a multistage counter (TC output connected to CET input), the delay t_{PHL} from CP to TC of a counter may exceed the delay t_{PLH} from CP to Q0 of the following (adjacent) counter. Under the conditions of the first circuit changing from count 15_{10} to 0 and the following (MSB) circuit changing from count 1_{10} to 15_{10}, the TC output of that MSB counter could generate an illegal logic "H" pulse.

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6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

6.5.3 Module crosstalk relationship. When connected in the worst case mode of operation (all but one module output switching simultaneously from a high level to a low level), the FDA module could exhibit substantial crosstalk. When the quiet line is a low level, there may be as much as a 1.2 volt positive going spike coincident with the negative edge of the switching outputs. The crosstalk spike must settle out to less than 400 mV after a delay of 50 ns maximum referenced from the positive going edge of the clock pulse.

6.6 Changes from previous issue. The margins of this specification are marked with an asterick to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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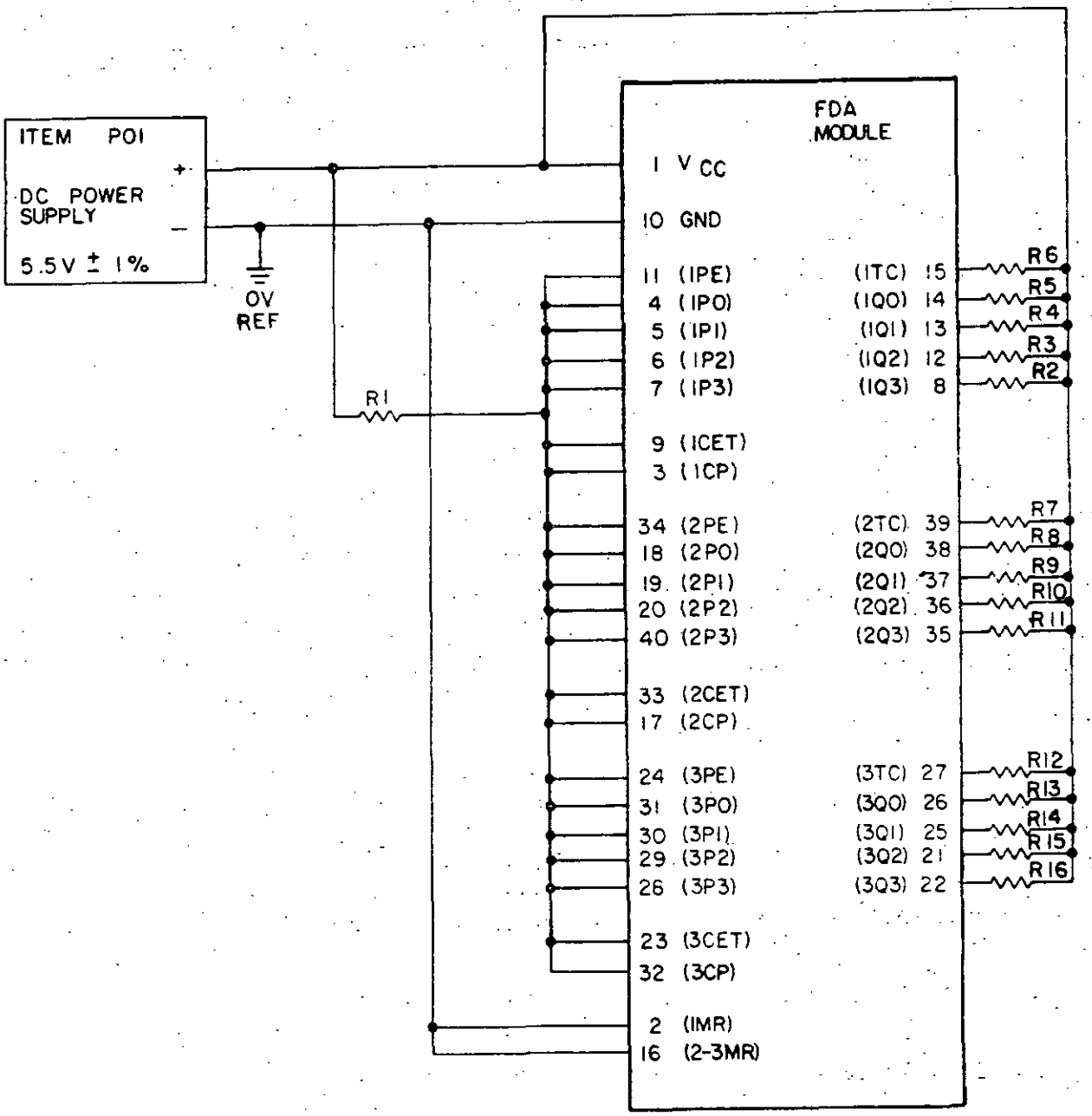
TABLE II. Input to output relationship. 1/-7/

Synchronous input before clock pulse						Required condition of asynchronous inputs	Outputs after clock pulse				
CET	PE	P3	P2	P1	P0	MR	TC	Q3	Q2	Q1	Q0
L	H	D3	D2	D1	D0	H	L	NC	NC	NC	NC
H	L	D3	D2	D1	D0	H	*	D3	D2	D1	D0
L	L	H	H	H	H	H	L	H	H	H	H
X	L	L	L	L	L	H	L	L	L	L	L
H	H	X	X	X	X	H	L	L	L	L	H
H	H	X	X	X	X	H	L	L	L	H	L
H	H	X	X	X	X	H	L	L	L	H	H
H	H	X	X	X	X	H	L	L	H	L	L
H	H	X	X	X	X	H	L	L	H	L	H
H	H	X	X	X	X	H	L	L	H	H	L
H	H	X	X	X	X	H	L	L	H	H	H
H	H	X	X	X	X	H	L	H	L	L	L
H	H	X	X	X	X	H	L	H	L	L	H
H	H	X	X	X	X	H	L	H	L	H	L
H	H	X	X	X	X	H	L	H	L	H	H
H	H	X	X	X	X	H	L	H	H	L	L
H	H	X	X	X	X	H	L	H	H	L	H
H	H	X	X	X	X	H	L	H	H	H	L
H	H	X	X	X	X	H	H	H	H	H	H
H	H	X	X	X	X	H	L	L	L	L	L

Asynchronous input		Required condition of synchronous inputs						Outputs				
MR	CET	CP	PE	P3	P2	P1	P0	TC	Q3	Q2	Q1	Q0
H	X	X	X	X	X	X	X	X	X	X	X	X
L	X	X	X	X	X	X	X	L	L	L	L	L
H	L	L	L	H	H	H	H	L	L	L	L	L
H	L	H	L	H	H	H	H	L	H	H	H	H
H	L	H	H	X	X	X	X	L	H	H	H	H
H	L	L	H	X	X	X	X	L	H	H	H	H
H	H	L	H	X	X	X	X	H	H	H	H	H
H	L	L	H	X	X	X	X	L	H	H	H	H

1/ TC (terminal count) = CET ·Q0·Q1·Q2·Q3.
2/ NC = No change from previous state.
3/ D0-D3 = Known input and output data (either high level or low level).
4/ X = Unspecified inputs or outputs (either high level or low level).
5/ H = High level.
6/ L = Low level.
7/ = If D0-D3 are "H" then TC is "H"; other TC is "L".

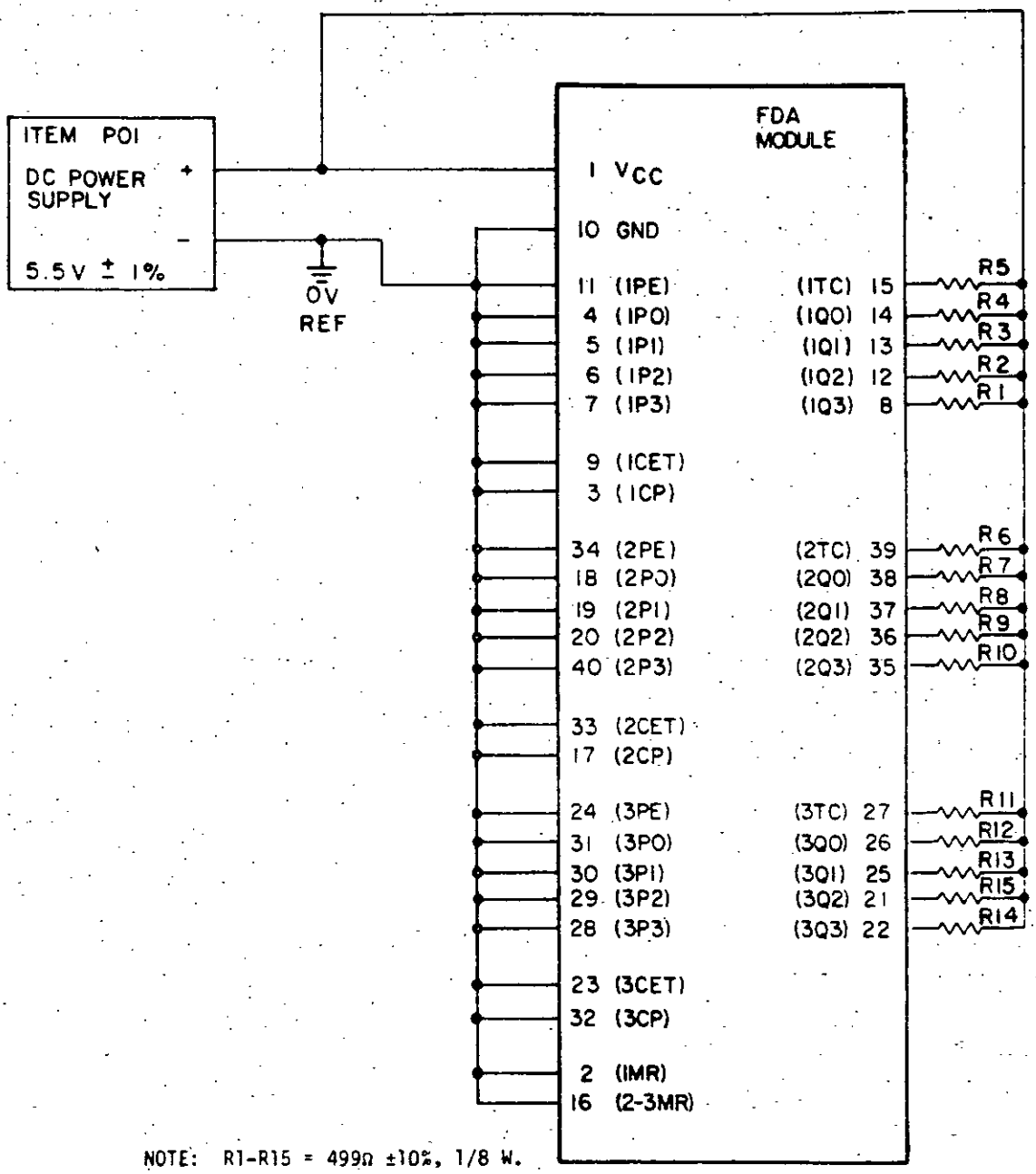
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NOTE: R1-R16 = 499Ω ±10%, 1/8 W.

FIGURE 2. Operating temperature and life test circuit.

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NOTE: R1-R15 = 499Ω ±10%, 1/8 W.

FIGURE 3. Transient thermal test circuit.

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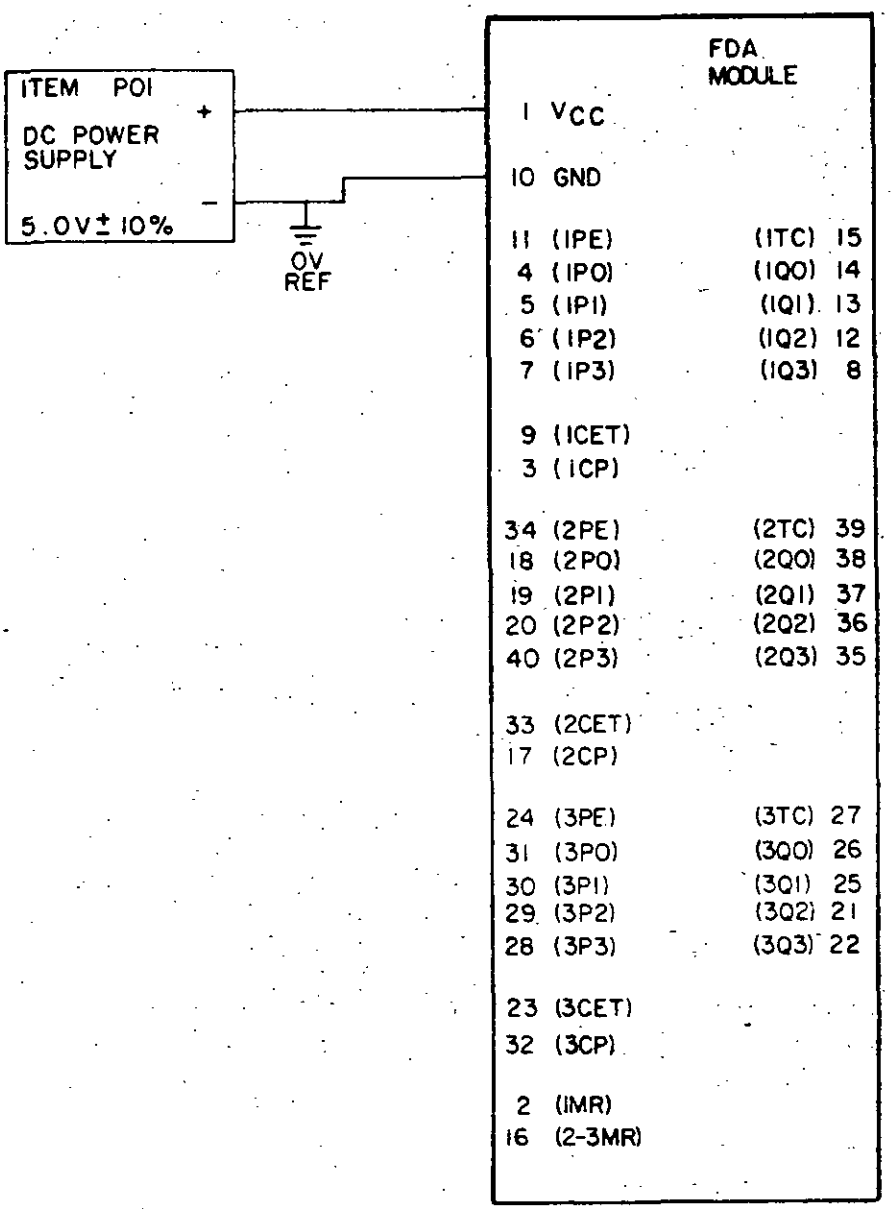


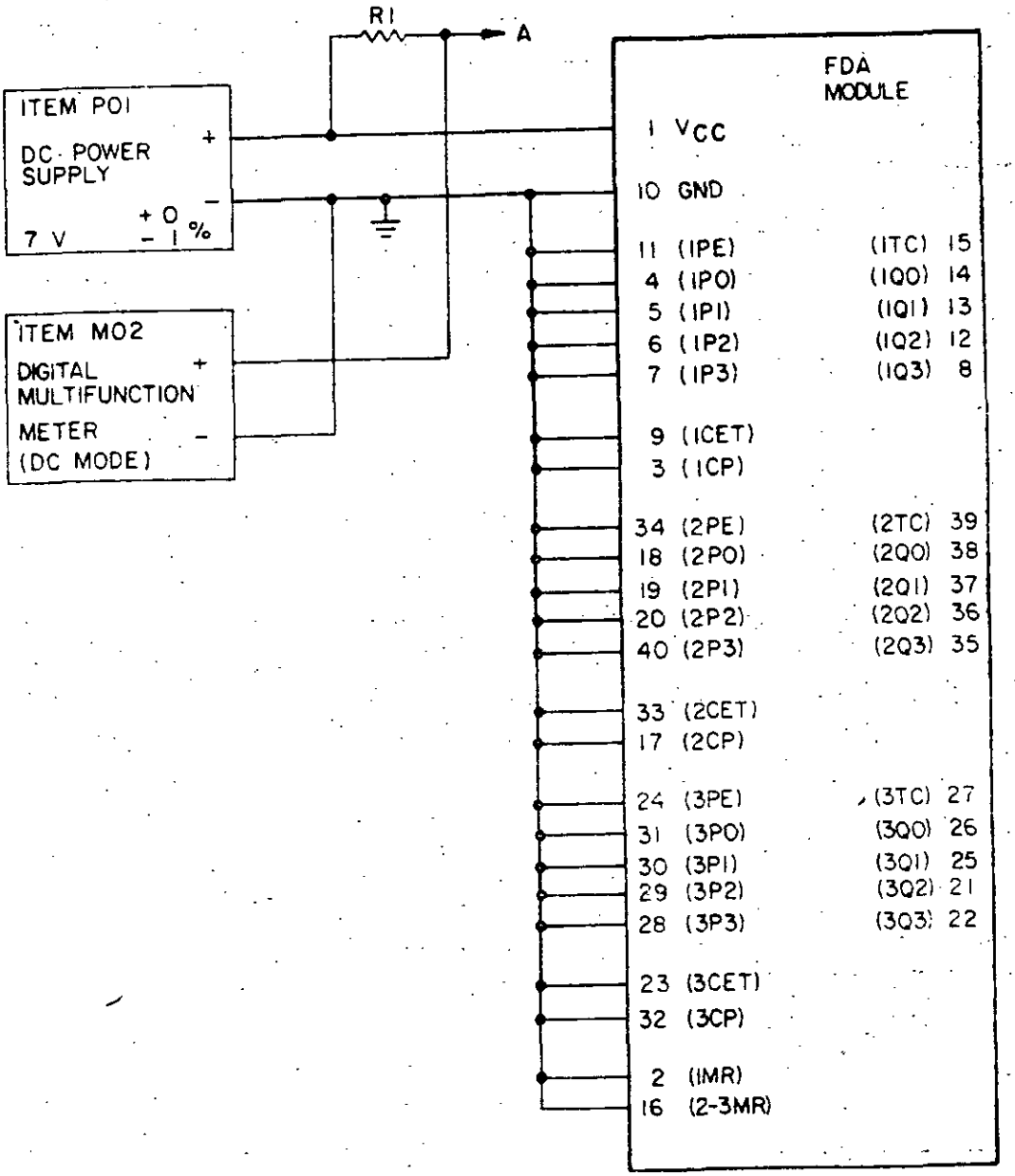
FIGURE 4. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
A1	1Q3	V_{OUT} (low loaded)	8	10	800	mV	5
A2	1Q2		12				
A3	1Q1		13				
A4	1Q0		14				
A5	1TC		15				
A6	2Q3		35				
A7	2Q2		36				
A8	2Q1		37				
A9	2Q0		38				
A10	2TC		39				
A11	3Q2		21				
A12	3Q3		22				
A13	3Q1		25				
A14	3Q0		26				
A15	3TC		27				

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NOTES:
1. R1 = 523Ω ±1%, 1/8 W.
2. After point A has been connected to the appropriate moduel pin, the output voltage shall be monitored for a minimum of 5 seconds before the measurement is recorded.

★ FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input breakdown current test.

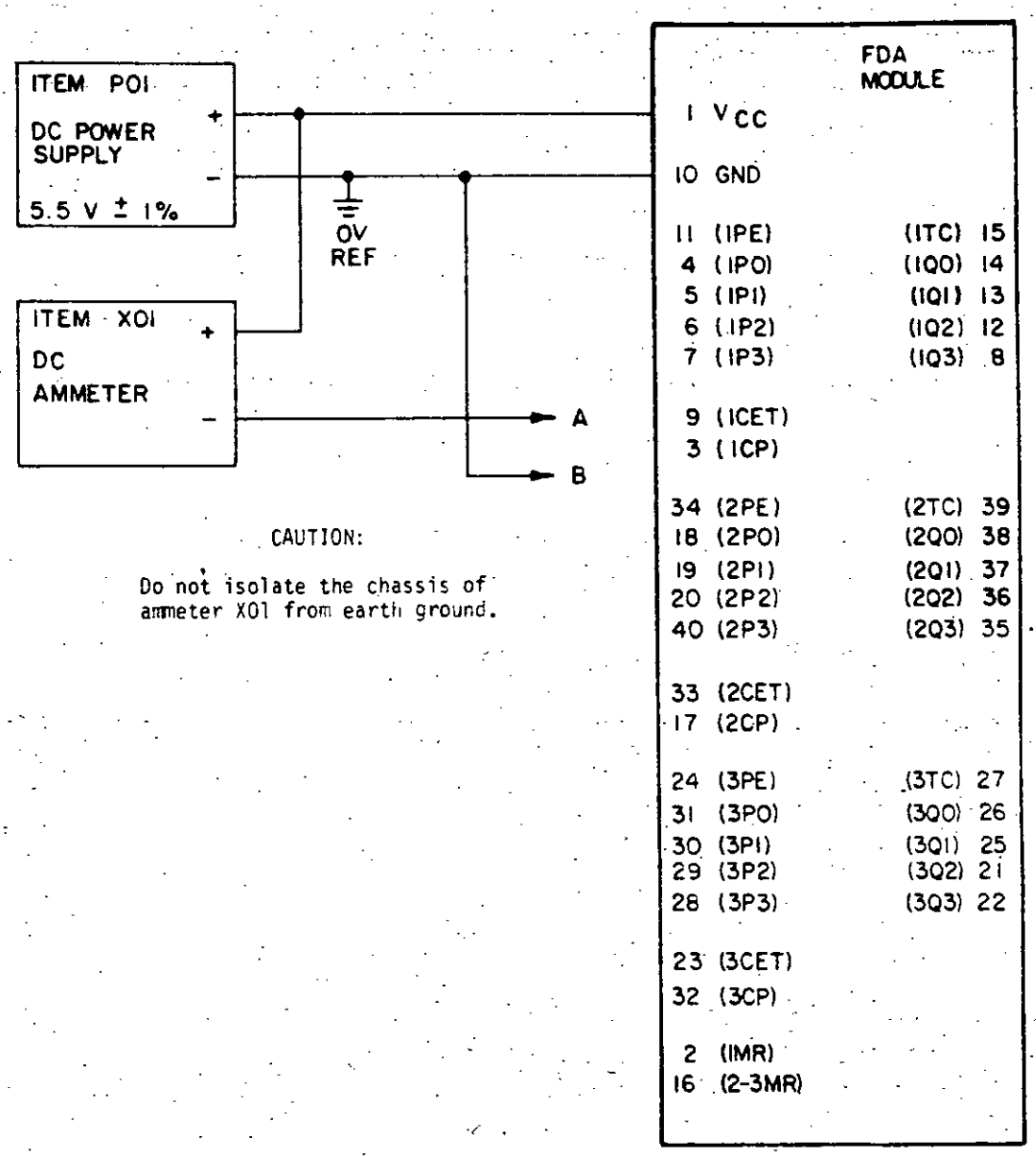
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	?-3MR Input breakdown current	16	17,18,19,20, 23,24,28,29, 30,31,32,33, 34 40	0	2	mA	6
B2	1CP	3	2,4,5,6,7,9, 11		1		
B3	1CET	9	2,3,4,5,6,7, 11				
B4	1PE	11	2,3,4,5,6,7,9				
B5	1MR	2	3,4,5,6,7,9 11				
B6	2CP	17	16,18,19,20, 33,34,40				
B7	2CET	33	16,17,18,19, 20,34,40				
B8	2PE	34	16,17,18,19, 20,33,40				
B9	3CET	23	16,24,28,29, 30,31,32				
B10	3PE	24	15,23,28,29, 30,31,32				
B11	3CP	32	16,23,24,28, 29,30,31				
B12	1PO	4	2,3,5,6,7,9, 11				

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TABLE IV. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B13	1P1	Input breakdown current	5	2,3,4,6,7,9, 11	0	1	mA	6
B14	1P2		6	2,3,4,5,7 9, 11				
B15	1P3		7	2,3,4,5,6,9, 11				
B16	2P0		18	16,17,19,20, 33,34,40				
B17	2P1		19	16,17,18,20, 33,34,40				
B18	2P2		20	16,17,18,19, 33,34,40				
B19	2P3		40	16,17,18,19, 20,33,34				
B20	3P3		28	16,23,24,29, 30,31,32				
B21	3P2		29	16,23,24,28, 30,31,32				
B22	3P1		30	16,23,24,28, 29,31,32				
B23	3P0		31	16,23,24,28, 29,30,32				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 6. Input breakdown current test circuit for table IV.

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TABLE V. Input clamp diode voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	1MR Input clamp diode voltage	2	-0.4	-1.5	V	7
C2	1CP	3				
C3	1PO	4				
C4	1P1	5				
C5	1P2	6				
C6	1P3	7				
C7	1CET	9				
C8	1PE	11				
C9	2-3MR	16				
C10	2CP	17				
C11	2PO	18				
C12	2P1	19				
C13	2P2	20				
C14	2CET	33				
C15	2PE	34				
C16	2P3	40				
C17	3CET	23				
C18	3PE	24				

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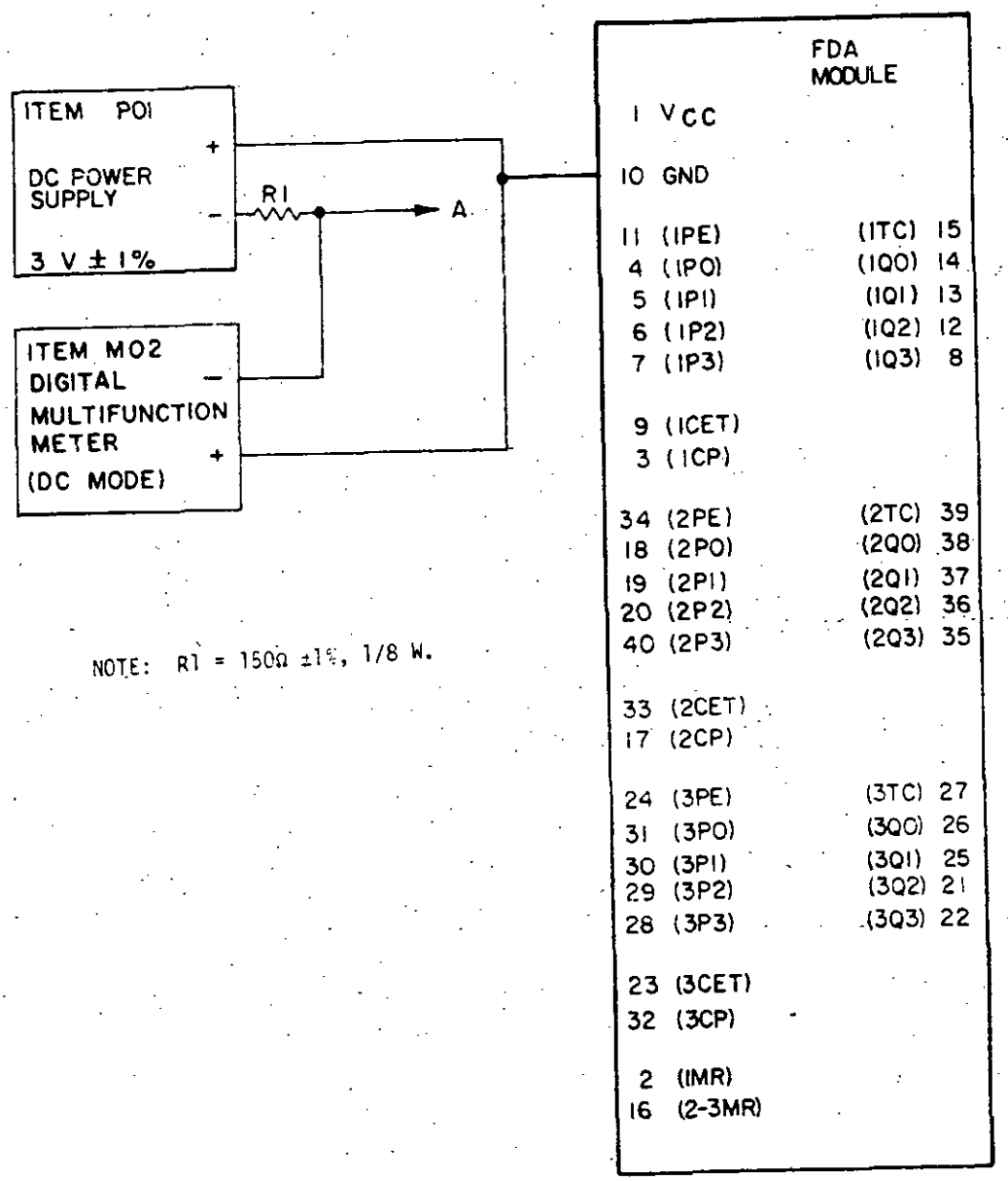
TABLE V. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C19	3P3	Input clamp diode voltage	28	-0.4	-1.5	V	7
C20	3P2	↓	29	↓	↓	↓	↓
C21	3P1		30				
C22	3P0		31				
C23	3CP		32				

TABLE VI. Isolation test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D	Isolation	All pins	Frame	10	∞	MΩ	8

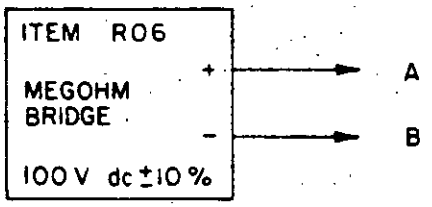
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NOTE: R1 = 150Ω ±1%, 1/8 W.

* FIGURE 7. Input clamp diode voltage test circuit for table V.

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FDA MODULE			
1	V_{CC}		
10	GND		
11	(1PE)	(1TC)	15
4	(1PO)	(1QO)	14
5	(1P1)	(1Q1)	13
6	(1P2)	(1Q2)	12
7	(1P3)	(1Q3)	8
9	(1CET)		
3	(1CP)		
34	(2PE)	(2TC)	39
18	(2PO)	(2QO)	38
19	(2P1)	(2Q1)	37
20	(2P2)	(2Q2)	36
40	(2P3)	(2Q3)	35
33	(2CET)		
17	(2CP)		
24	(3PE)	(3TC)	27
31	(3PO)	(3QO)	26
30	(3P1)	(3Q1)	25
29	(3P2)	(3Q2)	21
28	(3P3)	(3Q3)	22
23	(3CET)		
32	(3CP)		
2	(1MR)		
16	(2-3MR)		

FIGURE 8. Isolation test circuit for table VI.

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* TABLE VII. Functional (high-level and low-level), functional and static output voltage (high-level and low-level), and input threshold check test.

Test condition	Parameter		Connect								Output logic states					Initial limits 25 ±5°C ambient		Unit	Test figure
			A	B	C	D	E	F	G	H	TC	Q3	Q2	Q1	Q0	Min	Max		
E1	1CET	Functional, high level	Open	4,5,6,7	9,11	Open	Open	3	8	2	H	H	H	H	H	---	---	---	9
E2	2CET	Functional, high level	Open	18,19, 20,40	33,34	Open		17		16	H	H	H	H	H	---	---	---	
E3	3CET	Functional, high level	Open	28,29, 30,31	23,24	Open		32		16	H	H	H	H	H	---	---	---	
F1	1CET	Functional, low level	4,5,6,7	G	11	9		3		2	L	L	L	L	L	---	---	---	
F2	1CET		4,5,6,7		Open	9,11	15	3		2	L	H	H	H	H	0	400	mV	
F3	2CET		18,19, 20,40		34	33	Open	17		16	L	L	L	L	L	---	---	---	
F4	2CET		18,19, 20,40		Open	33,34	39	17			L	H	H	H	H	0	400	mV	
F5	3CET		28,29, 30,31		24	23	Open	32			L	L	L	L	L	---	---	---	
F6	3CET		28,29, 30,31		Open	23,24	27	32			L	H	H	H	H	0	400	mV	
G1	1TC	Functional and static output voltage, high, level	Open		4,5,6, 7,9	11	15	3		2	H	H	H	H	H	2.4	4.5	V	
G2	1Q3				7	4,5,6, 9,11	8				L	H	L	L	L				
G3	1Q2				6	4,5,7, 9,11	12				L	L	H	L	L				
G4	1Q1				5	4,6,7, 9,11	13				L	L	L	H	L				

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TABLE VII. Functional (high-level and low-level), functional and static output voltage (high-level and low-level), and input threshold check test - Continued.

Test condition	Parameter		Connect								Output logic states					Initial limits 25°±5°C ambient		Unit	Test figure
			A	B	C	D	E	F	G	H	TC	Q3	Q2	Q1	Q0	Min	Max		
G5	1Q0	Functional and static output voltage, high level	Open	G	4	5,6,7, 9,11	14	3	B	2	L	L	L	L	H	2.4	4.5	V	9
G6	2TC				18,19, 20,33, 40	34	39	17		16	H	H	H	H	H				
G7	2Q3				40	18,19, 20,33, 34	35				L	H	L	L	L				
G8	2Q2				20	18,19, 33,34, 40	36				L	L	H	L	L				
G9	2Q1				19	18,20, 33,34, 40	37				L	L	L	H	L				
G10	2Q0				18	19,20, 33,34, 40	38				L	L	L	L	H				
G11	3TC				23,28, 29,30, 31	24	27	32			H	H	H	H	H				
G12	3Q3				28	23,24, 29,30, 31	22	32			L	H	L	L	L				
G13	3Q2				29	23,24, 28,30, 31	21	32			L	L	H	L	L				

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* TABLE VII. Functional (high-level and low-level), functional and static output voltage (high-level and low-level), and input threshold check test - Continued.

Test condition	Parameter		Connect								Output logic states					Initial limits 25°±5°C ambient		Unit	Test figure
			A	B	C	D	E	F	G	H	TC	Q3	Q2	Q1	Q0	Min	Max		
G14	3Q1	Functional and static output voltage, high level	Open	G	30	23,24, 28,29, 31	25	32	B	16	L	L	L	H	L	2.4	4.5	V	9
G15	3Q0	Functional and static output voltage, high level			31	23,24, 28,29, 30	26	32		16		L	L	L	H	2.4	4.5	V	
H1	1Q3	Functional and static output voltage, low level			4,5,6	7,9,11	8	3		2		L	H	H	H	0	400	mV	
H2	1Q2				4,5,7	6,9,11	12					H	L	H	H				
H3	1Q1				4,6,7	5,9,11	13					H	H	L	H				
H4	1Q0				5,6,7	4,9,11	14					H	H	H	L				
H5	2Q3				18,19, 20	40,33, 34	35	17		16		L	H	H	H				
H6	2Q2				18,19, 40	20,33, 34	36					H	L	H	H				
H7	2Q1				18,20, 40	19,33, 34	37					H	H	L	H				
H8	2Q0				19,20, 40	18,33, 34	38					H	H	H	L				
H9	3Q3				29,30, 31	23,24, 28	22	32				L	H	H	H				
H10	3Q2				28,30, 31	23,24, 29	21	32				H	L	H	H				

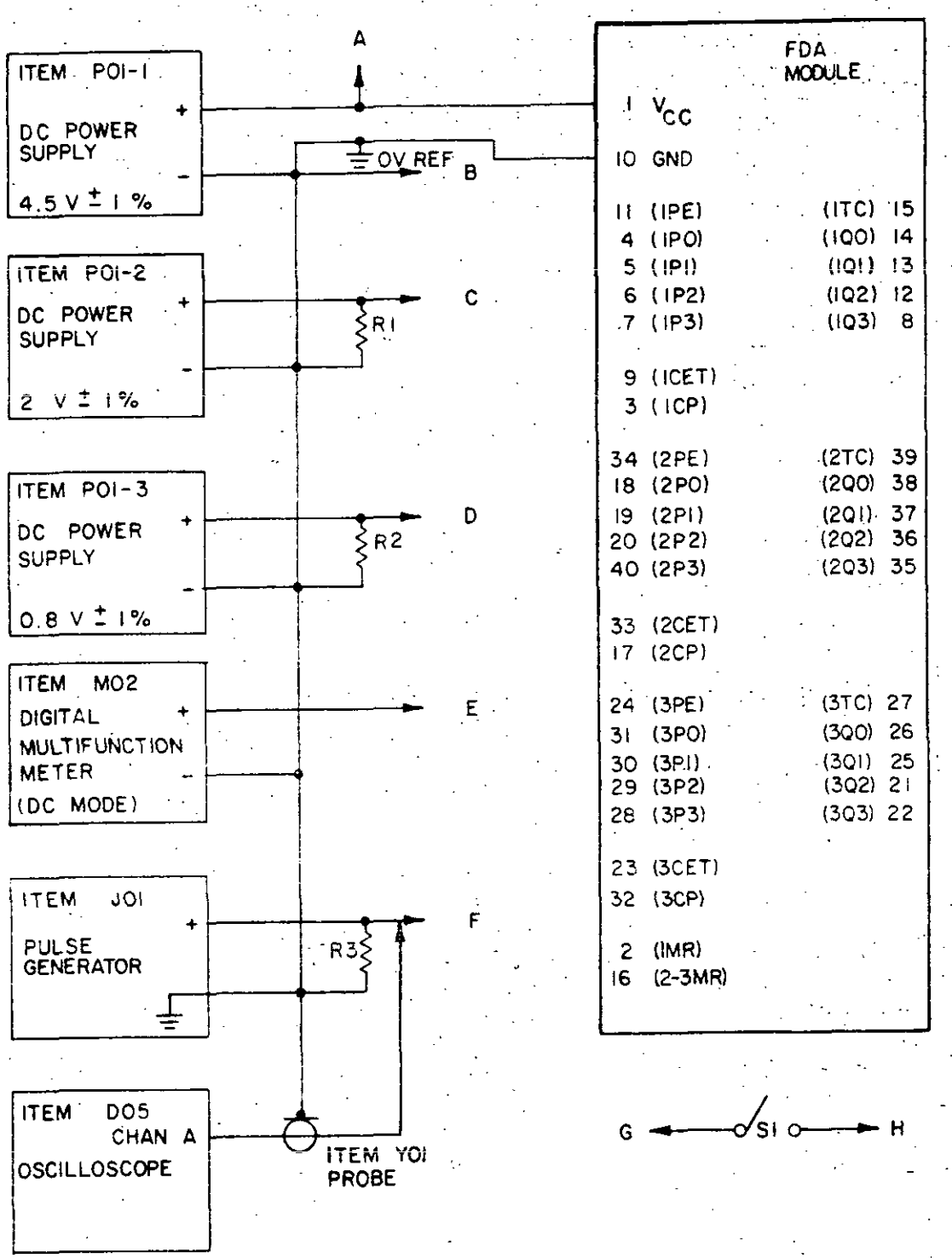
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* TABLE VII. Functional (high-level and low-level), functional and static output voltage (high-level and low-level), and input threshold check test - Continued.

Test condition	Parameter		Connect								Output logic states					Initial limits 25°±5°C ambient		Unit	Test figure
			A	B	C	D	E	F	G	H	TC	Q3	Q2	Q1	Q0	Min	Max		
H11	3Q1	Functional and static output voltage, low level	Open	G	28,29, 31	23,24, 30	25	32	B	16	L	H	H	L	H	0	400	mV	9
H12	3Q0	Functional and static output voltage, low level	Open	G	28,29, 30	23,24, 31	26	32	B	16	L	H	H	H	L	0	400	mV	
J1	1MR	Input threshold check	4,5,6, 7,9	11	Open	G	Open	2	D	3	L	L	L	L	L	---	---	---	
J2	2MR		18,19, 20,33, 40	34	Open	G		16	D	17	L	L	L	L	L	---	---	---	
J3	3MR		23,28, 29,30, 31	24	Open	G		16	D	32	L	L	L	L	L	---	---	---	
K1	1CET		Open	4,5, 6,7	9,11	Open		3	B	2	L	L	L	L	H	---	---	---	
K2	2CET		Open	18,19, 20,40	33,34	Open		17	B	16	L	L	L	L	H	---	---	---	
K3	3CET		Open	28,29, 30,31	23,24	Open		32	B	16	L	L	L	L	H	---	---	---	

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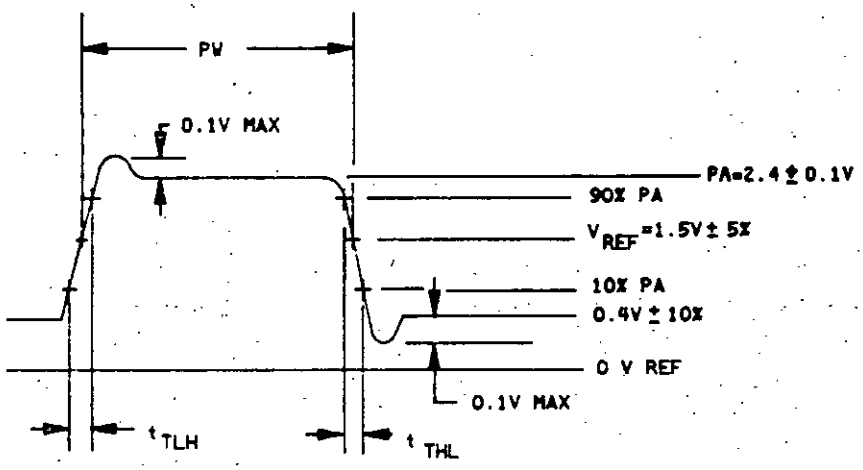
* FIGURE 9. Functional (high-level and low-level), functional and static output voltage (high-level and low-level), and input threshold check test circuit for table VII.

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NOTES:
1. R1 = R2 = R3 = 50Ω ±5%, 1 W.
2. S1 = SPST switch.
3. Attach an output logic load circuit to pins 8, 12, 13, 14, 15, 21, 22, 25, 26, 27, 35, 36, 37, 38, and 39. Each output logic load circuit shall be as specified on figure 10.
4. Adjust the item J01 input pulse width, transition times, and pulse amplitude at a pulse repetition rate of 30 kHz ±50% using item D05. Adjust the dc logic level with item J01 in the single pulse mode using item M02.
5. Test procedure for test conditions E1-E3:

   a. Input pulse at point F with F connected:

      PW = 100 ns ±10%.
      $t_{TLH} = t_{THL}$ = 20 ns ±10%.



   b. After making all circuit connections for a test condition close S1 for at least 100 ns then open S1. With item J01 in the single pulse mode apply 15 pulses to the circuit under test. The output logic states shall be as specified in table VII. The maximum pulse repetition rate shall be 1 kHz.

6. Test procedure for test conditions F1-F6:

   a. The input pulse at point F with F connected shall be as specified in note 5(a) above.
   b. After making all circuit connections for a test condition, close S1 for at least 100 ns then open S1. With item J01 in the single pulse mode, apply one pulse to the circuit under test. The output logic states and voltage levels shall be as specified in table VII.

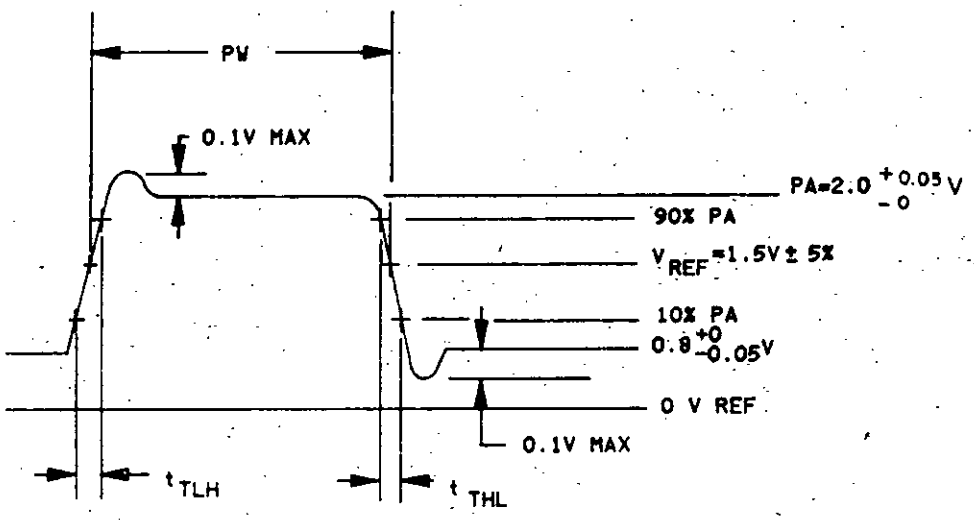
* FIGURE 9. Functional (high-level and low-level), functional and static output voltage (high-level and low-level), and input threshold check test circuit for table VII - Continued.

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7. Test procedure for test conditions G1-G15 and H1-H12:

a. Input pulse at point F with F connected:

PW = 100 ns ±10%.
$t_{TLH} = t_{THL}$ = 20 ns ±10%.



b. After making all circuit connections for a test condition, close S1 for at least 100 ns then open S1. With item J01 in the single pulse mode, apply one pulse to the circuit under test. The output logic states and voltage levels shall be as specified in table VII.

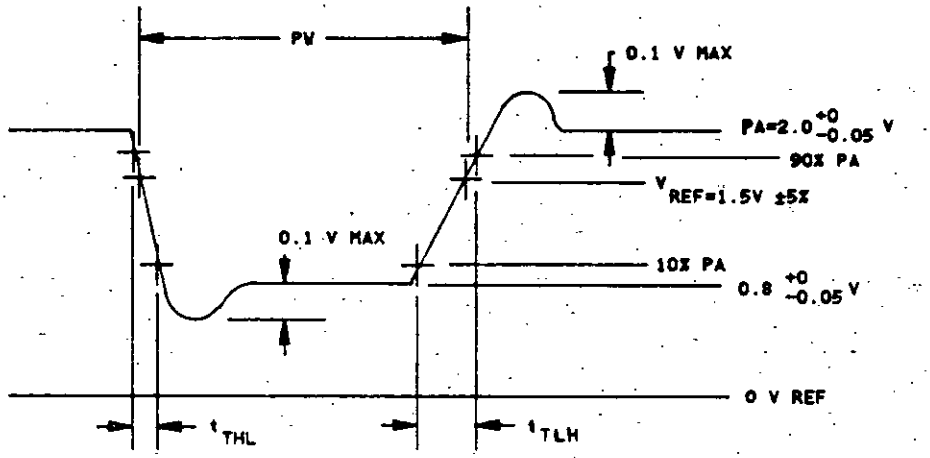
* FIGURE 9. Functional (high-level and low-level), functional and static output voltage (high-level and low-level), and input threshold check test circuit for table VII - Continued.

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8. Test procedure for test conditions J1-J3:

   a. Input pulse at point F with F connected:

      PW = 100 ns ±10%.
      $t_{TLH} = t_{THL}$ = 20 ns ±10%.



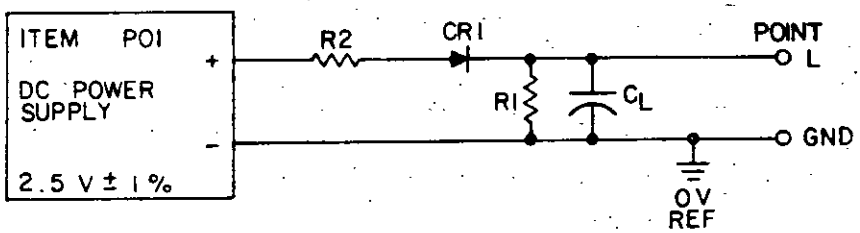
   b. Initially S1 shall be closed. After making all circuit connections for a test condition, open S1 for at least 100 ns then close S1. With item J01 in the single pulse mode, apply one pulse to the circuit under test. The output logic states shall be as specified in table VII.

9. Test procedure for test conditions K1-K3:

   a. The input pulse at point F with F connected shall be as specified in note 7(a) above.
   b. After making all circuit connections for a test condition, close S1 for at least 100 ns then open S1. With item J01 in the single pulse mode, apply one pulse to the circuit under test. The output logic states shall be as specified in table VII.

* FIGURE 9. Functional (high-level and low-level), functional and static output voltage (high-level and low-level), and input threshold check test circuit for table VII - Continued.

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NOTES:
1. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions S, T, U, V, W, X, Y, and Z.
2. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions GG and HH.
3. CR1 = 1N4150, or equivalent.
4. R1 = 6.65 kΩ ±1%, 1/8 W.
5. R2 = Select value for a current flow of 10 mA ±1% from point "L" to ground.
6. C_L is not required for dc electrical test conditions E, F, G, H, J, and K.
7. The maximum lead length from point "L" to the appropriate module pin shall be 2 inches (50.8 mm).
8. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

* FIGURE 10. Logic load circuit.

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TABLE VIII. Input current (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	1MR Input current, low level	Open	2	Open	Open	-0.1	-1.73	mA	11
L2	1CP		3				-3.46		
L3	1CET		9						
L4	1PE		11						
L5	2-3MR		16						
L6	2CP		17						
L7	2CET		33						
L8	2PE		34						
L9	3CET		23						
L10	3PE		24						
L11	3CP		32						
L12	1PO		4				-1.15		
L13	1P1		5						
L14	1P2		6						
L15	1P3		7						
L16	2PO		18						
L17	2P1		19						
L18	2P2		20						
L19	2P3		40						

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TABLE VIII. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L20	3P3 Input current, low level	Open	28	Open	Open	-0.1	-1.15	mA	11
L21	3P2		29						
L22	3P1		30						
L23	3PO		31						
M1	1MR Input current, high level	2	Open		3,4,5, 6,7,9, 11	0	40	μA	
M2	1CP	3			2,4,5, 6,7,9, 11		80		
M3	1CET	9			2,3,4, 5,6,7, 11				
M4	1PE	11			2,3,4, 5,6,7, 9				
M5	2-3MR	16			17,18, 19,20, 23,24, 28,29, 30,31, 32,33, 34,40				
M6	2CP	17			16,18, 19,20, 33,34, 40				

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TABLE VIII. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M7	2CET Input current, high level	33	Open	Open	16,17, 18,19, 20,34, 40	0	80	µA	11
M8	2PE	34			16,17, 18,19, 20,33, 40				
M9	3CET	23			16,24, 28,29, 30,31, 32				
M10	3PE	24			16,23, 28,29, 30,31, 32				
M11	3CP	32			16,23, 24,28, 29,30, 31				
M12	1PO	4			2,3,5, 6,7,9, 11		30		
M13	1P1	5			2,3,4, 6,7,9, 11		30		
M14	1P2	6			2,3,4, 6,7,9, 11		30		

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TABLE VIII. Input current (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
M15	1P3	Input current, high level	7	Open	Open	2,3,4, 5,6,9, 11	0	30	µA	11
M16	2P0		18			16,17, 19,20, 33,34, 40				
M17	2P1		19			16,17, 18,20, 33,34, 40				
M18	2P2		20			16,17, 18,19, 33,34, 40				
M19	2P3		40			16,17, 18,19, 20,33, 34				
M20	3P3		28			16,23, 24,29, 30,31, 32				
M21	3P2		29			16,23, 24,28, 30,31, 32				

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TABLE VIII. Input current (low-level and high-level) test - Continued.

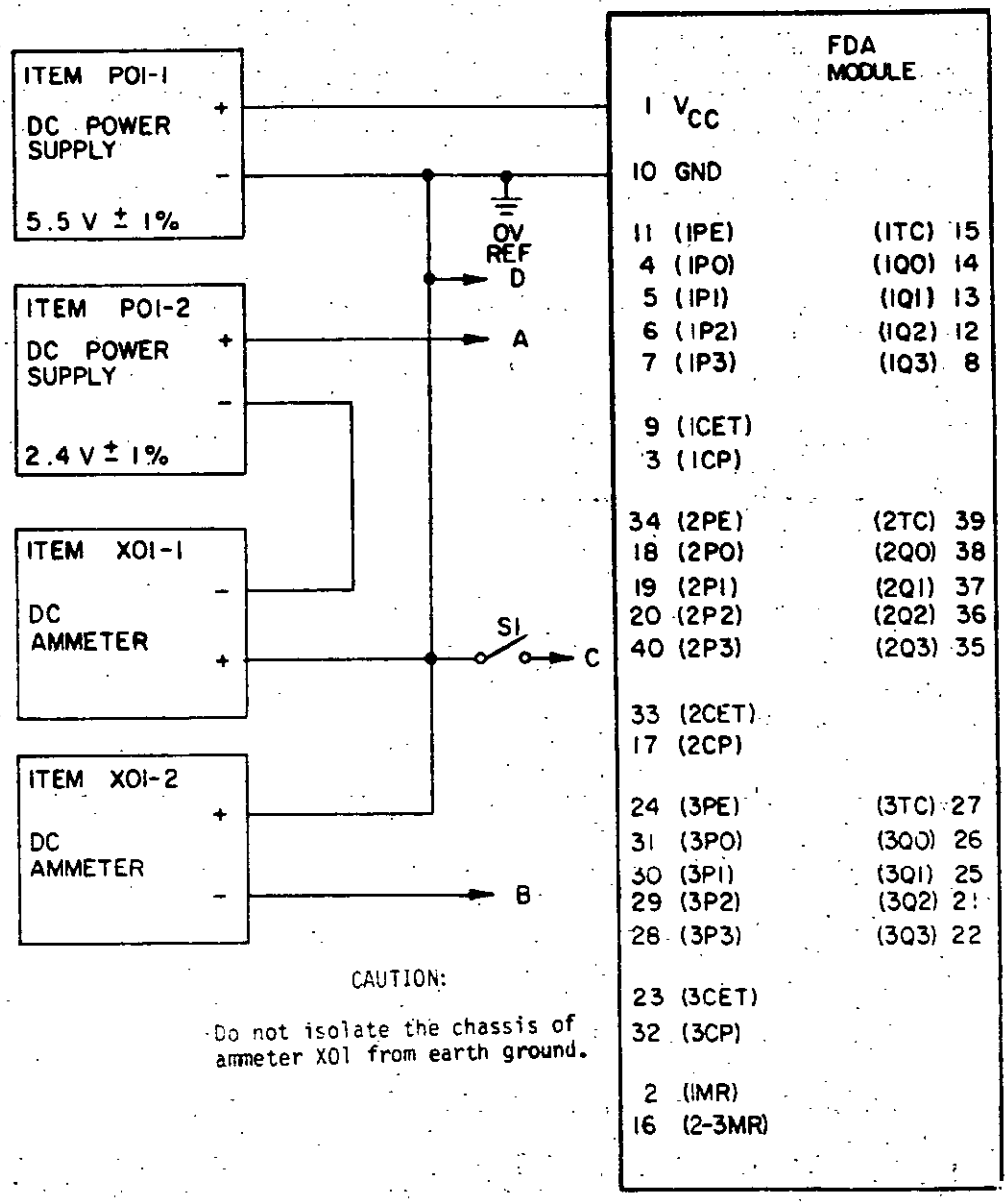
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M22	3P1 Input current, high level	30	Open	Open	16,23, 24,28, 29,31, 32	0	30	µA	11
M23	3P0 Input current, high level	31	Open	Open	16,23, 24,28, 29,30, 32	0	30	µA	11

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TABLE IX. Output short circuit current test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1Q3	Output short circuit current	Open	8	3,17,32	11,24,34	-20	-80	mA	11
N2	1Q2			12						
N3	1Q1			13						
N4	1Q0			14						
N5	1TC			15						
N6	2Q3			35						
N7	2Q2			36						
N8	2Q1			37						
N9	2Q0			38						
N10	2TC			39						
N11	3Q2			21						
N12	3T3			22						
N13	3Q1			25						
N14	3Q0			26						
N15	3TC			27						

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NOTES:
1. S1 = SPST switch.
2. Since the items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
3. After making all circuit connections for a test condition, except point A, close S1 for at least 100 ns then open S1. Then connect point A and take measurement.
4. Only one output shall be shorted at a time and the measurement taken within 10 seconds.

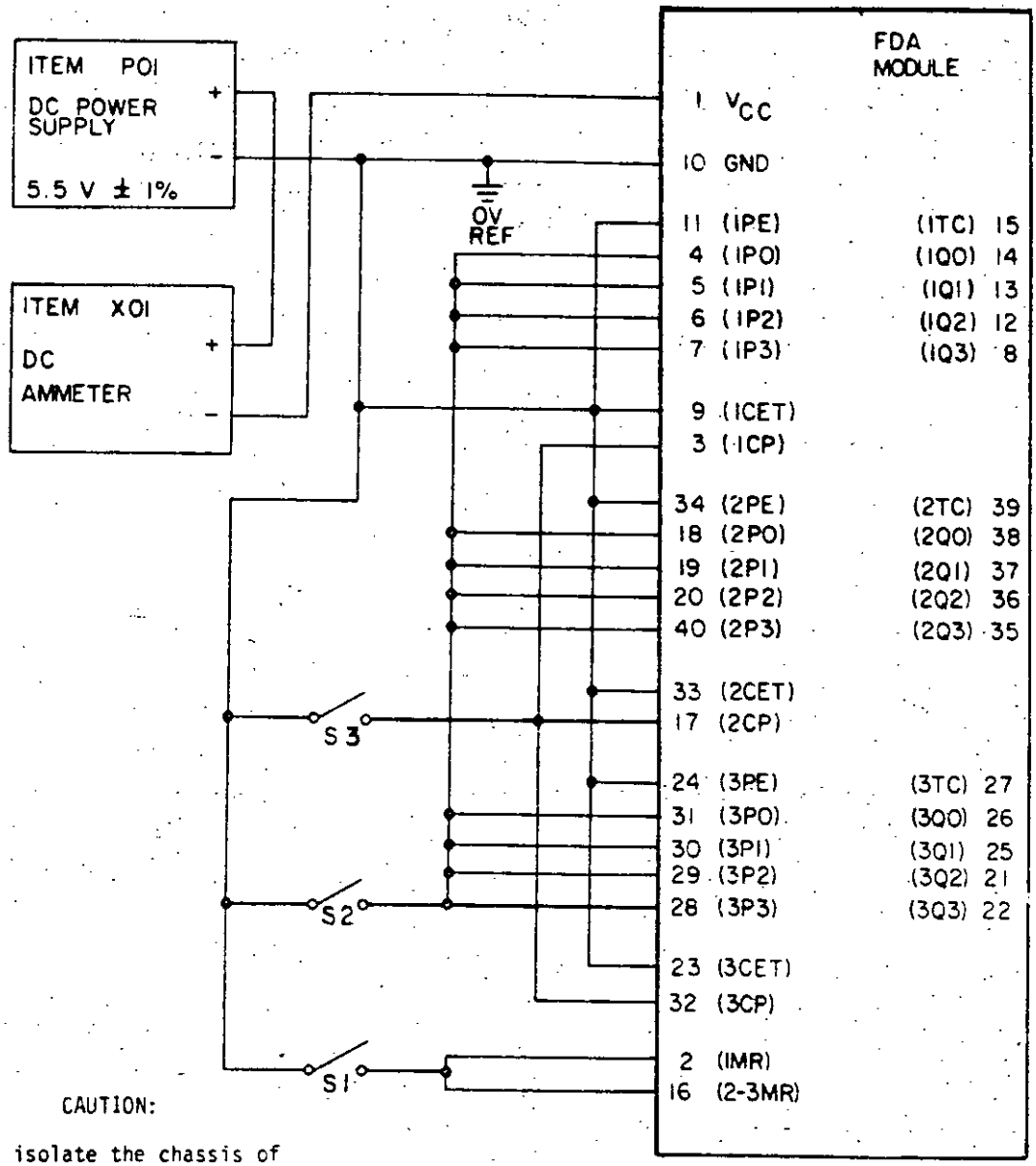
* FIGURE 11. Input current (low-level and high-level) and output short circuit current test circuit for tables VIII and IX.

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TABLE X. Power supply current (high-level and low-level) test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
P1	Power supply current, high level	50	290	mA	12
P2	Power supply current, low level	50	290	mA	12

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. S1 = S2 = S3 = SPST switch.
2. Test procedure for test condition P1:

   a. S1 open.
   b. Initially S2 shall be open and S3 closed.
   c. Open S3 for 100 ns minimum then close S3 and S2.
   d. Take measurement.

3. Test procedure for test condition P2:

   a. Close S1, S2, and S3.
   b. Take measurement.

FIGURE 12. Power supply current (high-level and low-level) test circuit for table X.

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TABLE XI. Input capacitance test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
Q1	2-3MR Input capacitance	16	2	30	pF	13
Q2	1MR	2		15		
Q3	1CP	3				
Q4	1PO	4				
Q5	1P1	5				
Q6	1P2	6				
Q7	1P3	7				
Q8	1CET	9				
Q9	1PE	11				
Q10	2CP	17				
Q11	2PO	18				
Q12	2P1	19				
Q13	2P2	20				
Q14	2CET	33				
Q15	2PE	34				
Q16	2P3	40				
Q17	2CET	23				
Q18	3PE	24				
Q19	3P3	28				

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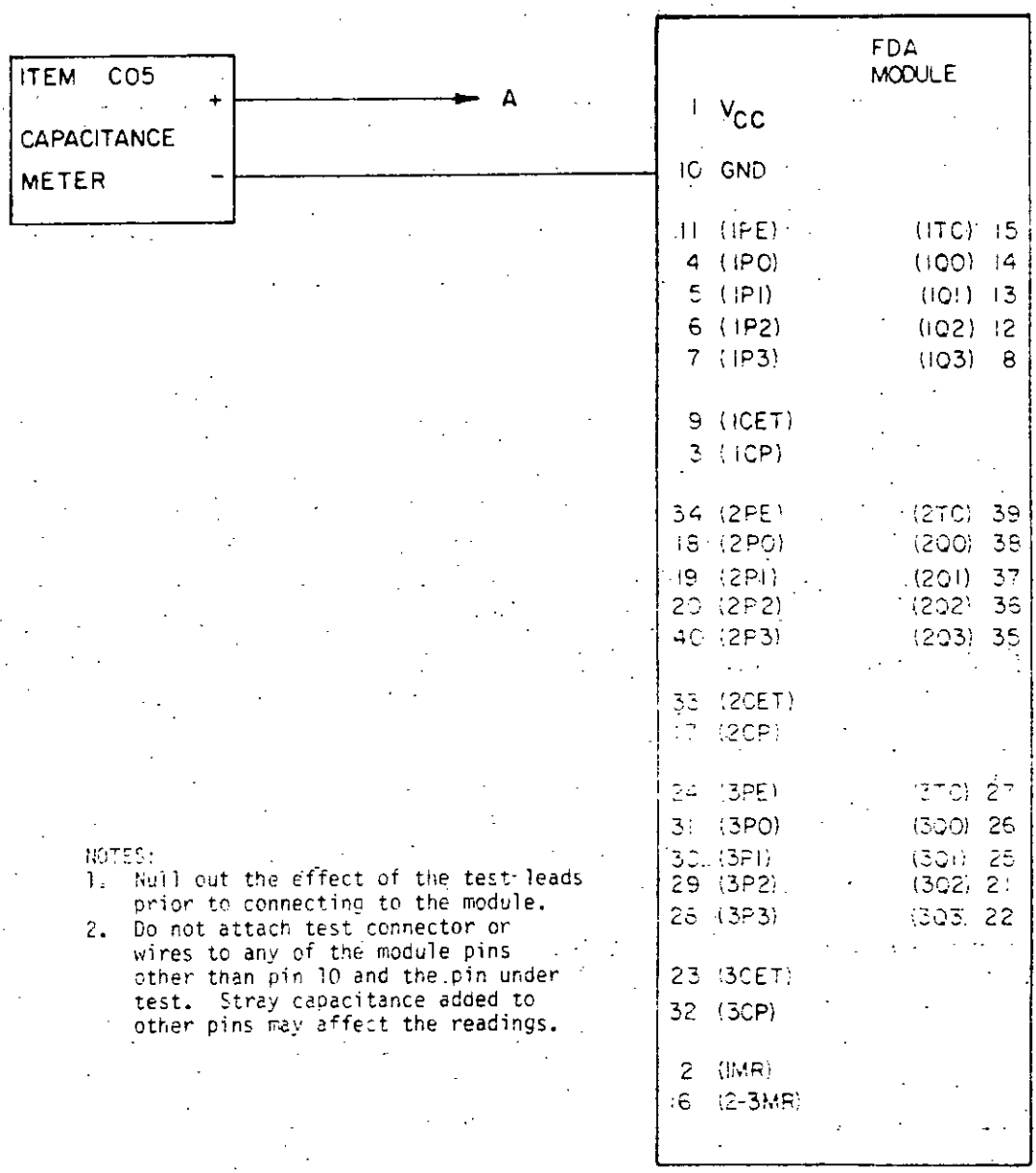
TABLE XI. Input capacitance test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
Q20	3P2 Input capacitance	29	2	15	pF	13
Q21	3P1	30	↓	↓	↓	↓
Q22	3P0	31				
Q23	3CP	32				

TABLE XII. Power supply transient amplitude (5 V) test.

Test condition	Parameter	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
R	Power supply transient amplitude (5 V)	0	400	mV	14,15

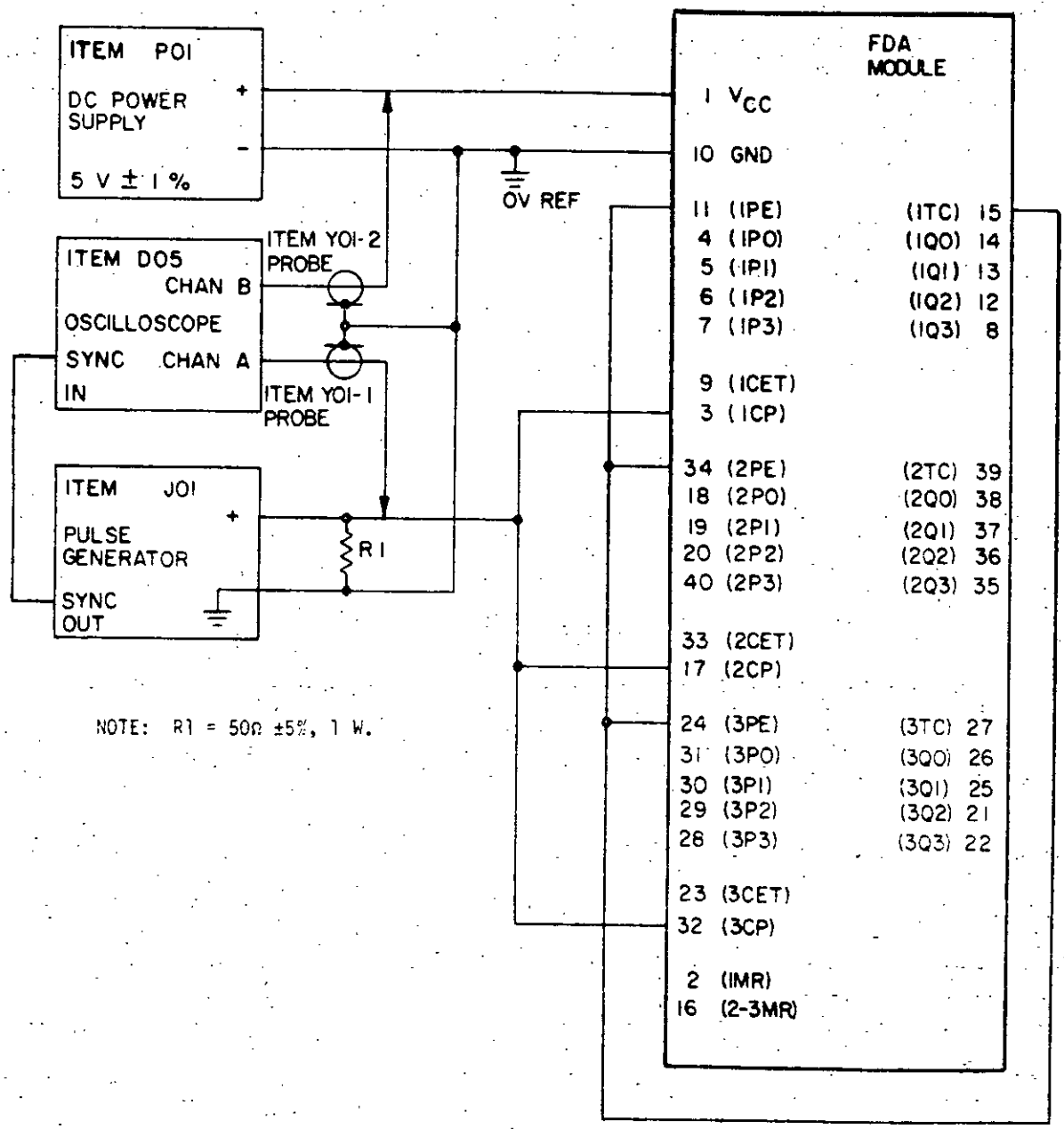
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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 13. Input capacitance test circuit for table XI.

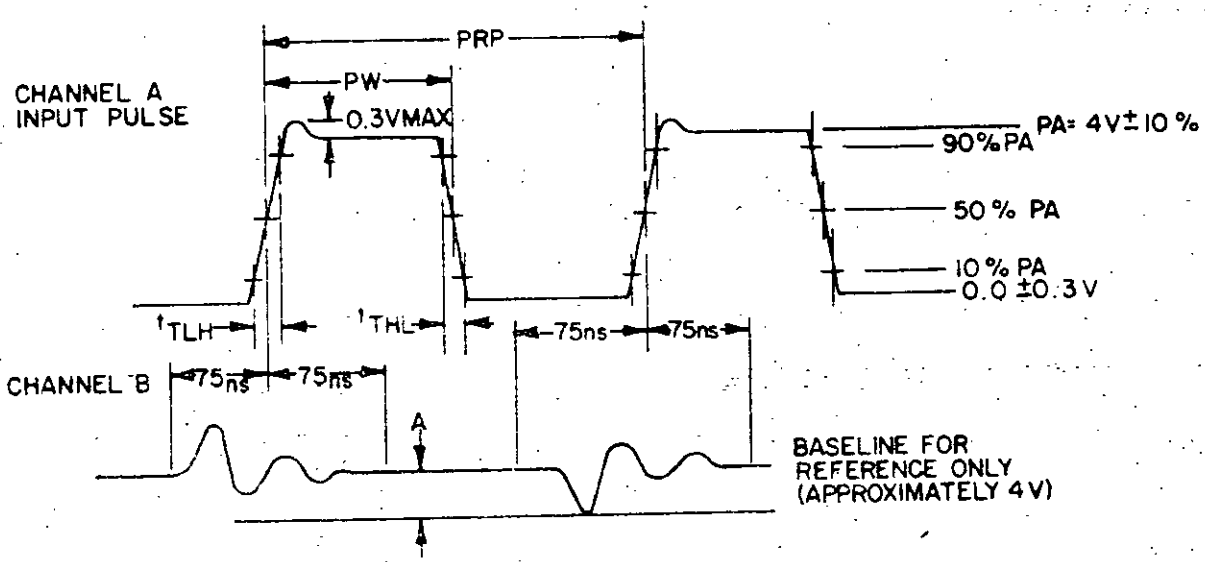
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NOTE: R1 = 50Ω ±5%, 1 W.

* FIGURE 14. Power supply transient amplitude (5 V) test circuit for table XII.

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NOTES:
1. Input pulse parameters:
   PW = 100 ns ±10%.
   PRP = 0.2 µs ±10% (adjusted under load).
   $t_{TLH} = t_{THL}$ = 10 ±0.2 ns.
2. "A" is measured from the baseline to the most negative excursion of the waveform, which may occur in time coincidence with either of alternate input pulses. Only negative excursions occurring within ±75 ns of the 50 percent point of the input waveform shall be considered.

* FIGURE 15. Power supply transient amplitude (5 V) test circuit waveforms for table XII.

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TABLE XIII. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PHL} and t_{PLH}) test.

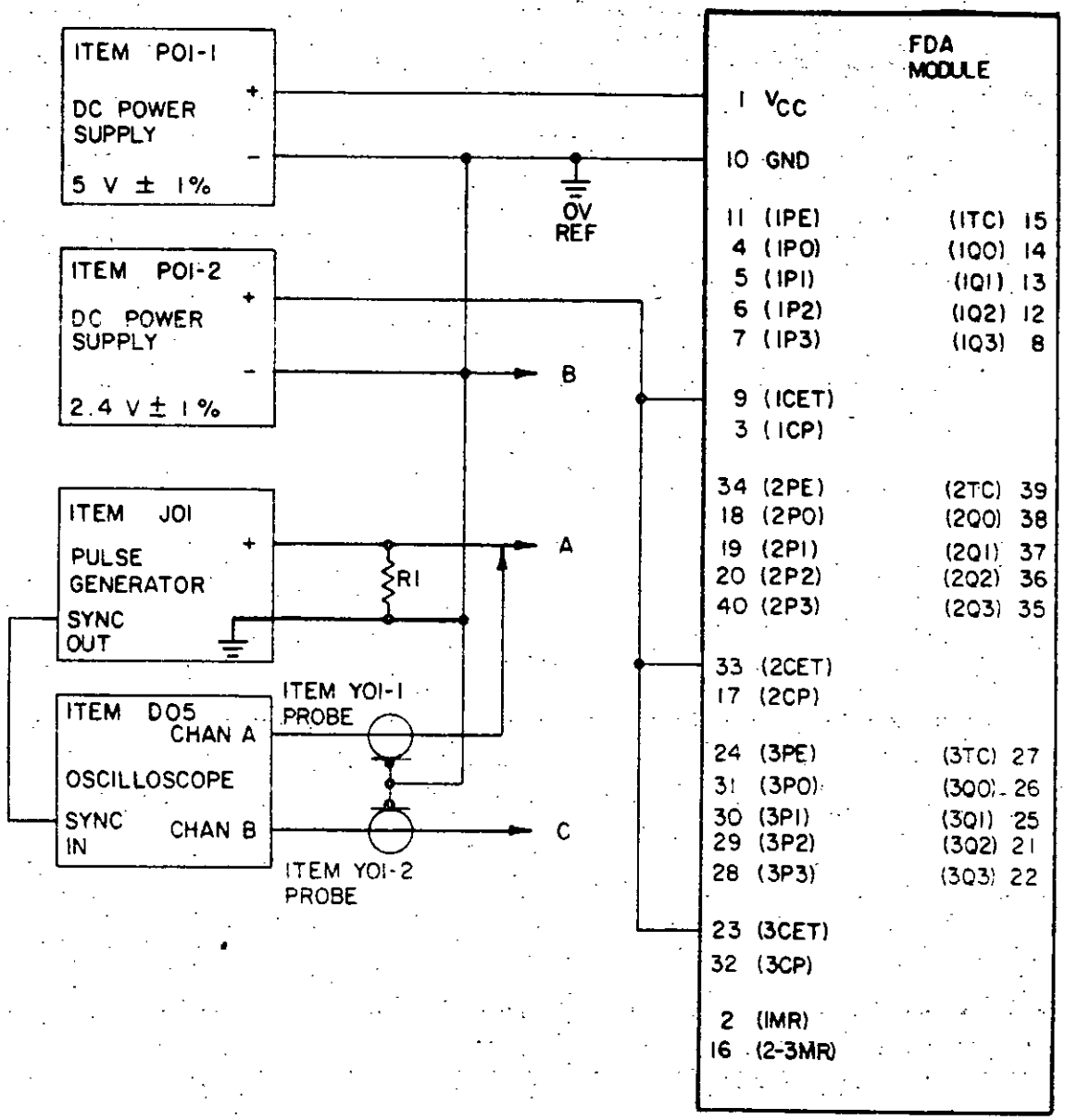
Test condition	Parameter	Connect					t_{THL} initial limits 25° ±5°C ambient		t_{PHL} initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin		Min	Max	Min	Max		
S1, U1	1TC Transition time (t_{THL}), propagation delay time (t_{PHL})	3	4	15	13	11	2	20	10	20	ns	16,17
S2, U2	1Q3	3	Open	8	13	11			2	45		
S3, U3	2TC	17	18	39	37	34			10	20		
S4, U4	2Q3	17	Open	35	37	34			2	45		
S5, U5	3TC	32	31	27	25	24			10	20		
S6, U6	3Q3	32	Open	22	25	24			2	45		
S7, U7	1Q0	3	6	14	13	11						
S8, U8	1Q1	3	6	13	14	11						
S9, U9	1Q2	3	7	12	13	11						
S10, U10	2Q0	17	20	38	37	34						
S11, U11	2Q1	17	20	37	38	34						
S12, U12	2Q2	17	40	36	37	34						
S13, U13	3Q0	32	29	26	25	24						
S14, U14	3Q1	32	29	25	26	24						
S15, U15	3Q2	32	28	21	25	24						

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TABLE XIII. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PHL} and t_{PLH}) test - Continued.

Test condition	Parameter	Connect					t_{TLH} initial limits 25° ±5°C ambient		t_{PLH} initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin		Min	Max	Min	Max		
T1, V1	1TC Transition time (t_{TLH}), propagation delay time (t_{PLH})	3	4	15	13	11	4	35	4	70	ns	16,17
T2, V2	2TC	17	18	39	37	34				70		
T3, V3	3TC	32	31	27	25	24				70		
T4, V4	1Q3	3	7	8	13	11				50		
T5, V5	2Q3	17	40	35	37	34				50		
T6, V6	3Q3	32	28	22	25	24				50		
T7, V7	1Q0	3	4	14	8	11			12	28		
T8, V8	1Q1	3	5	13	14	11			4	50		
T9, V9	1Q2	3	6	12	13	11			4	50		
T10, V10	2Q0	17	18	38	35	34			12	28		
T11, V11	2Q1	17	19	37	38	34			4	50		
T12, V12	2Q2	17	20	36	37	34			4	50		
T13, V13	3Q0	32	31	26	22	24			12	28		
T14, V14	3Q1	32	30	25	26	24			4	50		
T15, V15	3Q2	32	29	21	25	24			4	50		

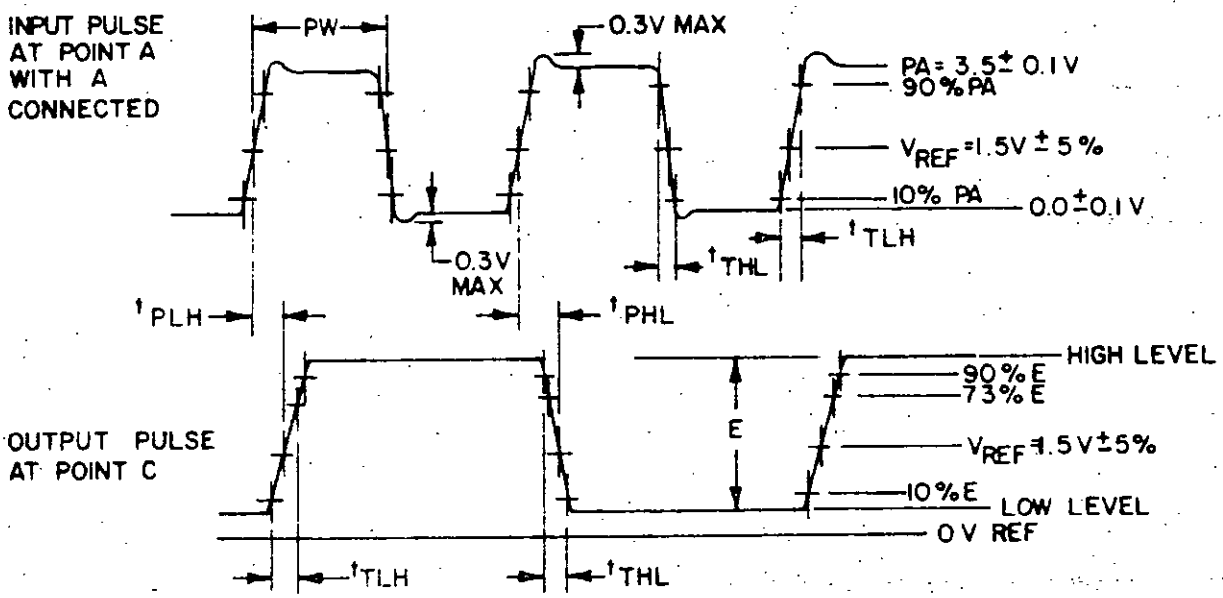
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NOTES:
1. RI = 50Ω ±5%, 1 W.
2. Load output pins 8, 12, 13, 14, 15, 21, 22, 25, 26, 27, 35, 36, 37, 38, and 39 with a load as specified on figure 10.

* FIGURE 16. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for table XIII.

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NOTES:
1. Input pulse parameters:
   PW = 100 ns ±10%
   PRR = 1 MHz ±10%
   $t_{TLH} = t_{THL} = 8 \pm 2$ ns
2. Before making the measurements verify that the output pulse low level is $\leq$ 0.4 V and the high level $\geq$ 2.4 V.

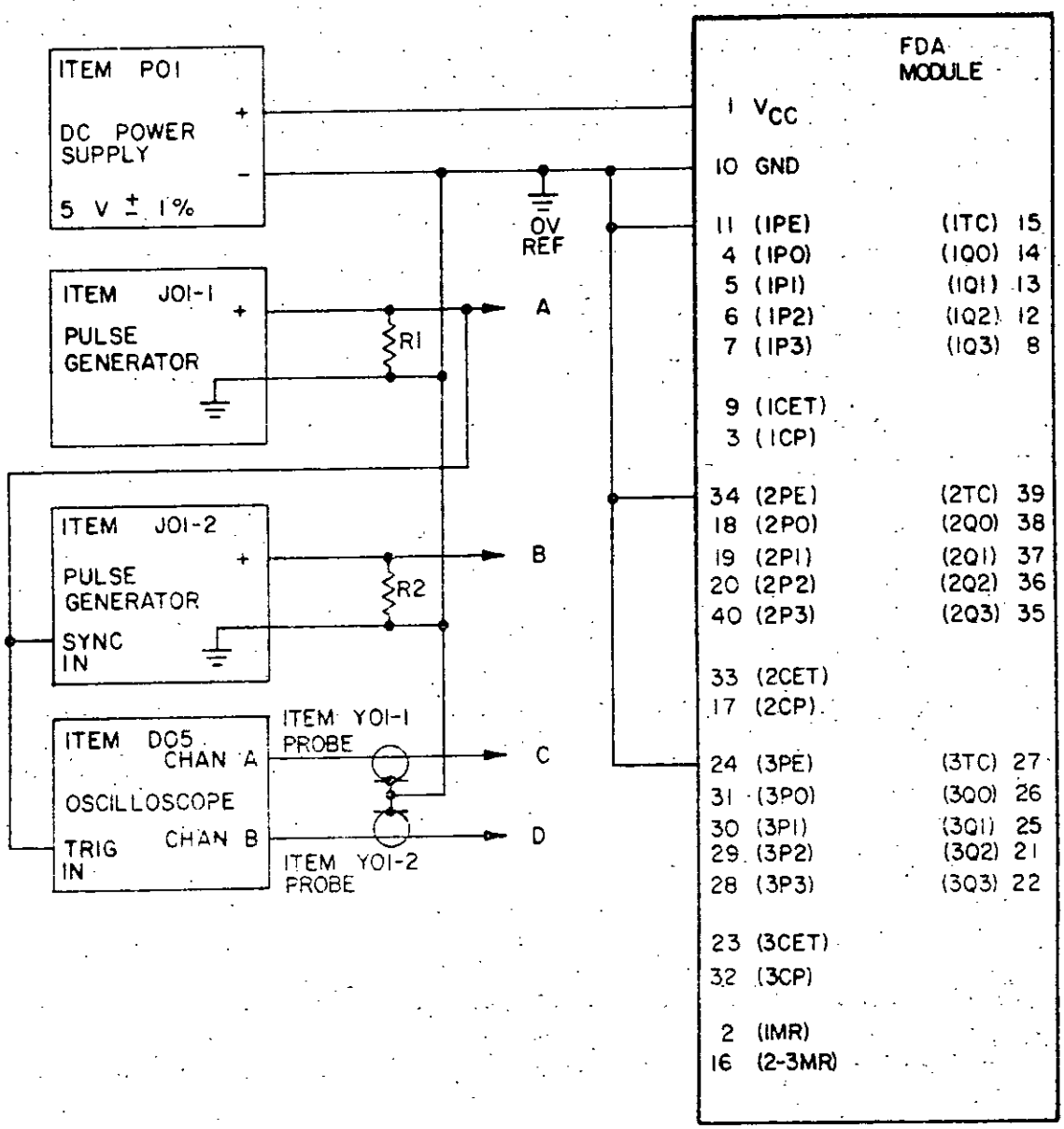
* FIGURE 17. Transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for table XIII.

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TABLE XIV. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
W1	1MR Propagation delay time (t_{PHL})	2	3	2	8	5	60	ns	18,19
W2	2-3MR	16	17	16	35				
W3	2-3MR	16	32	16	22				
W4	1MR	2	3	2	15				
W5	1MR				14				
W6	1MR				13				
W7	1MR				12				
W8	2-3MR	16	17	16	39				
W9	2-3MR				38				
W10	2-3MR				37				
W11	2-3MR				36				
W12	2-3MR		32		27				
W13	2-3MR				26				
W14	2-3MR				25				
W15	2-3MR				21				

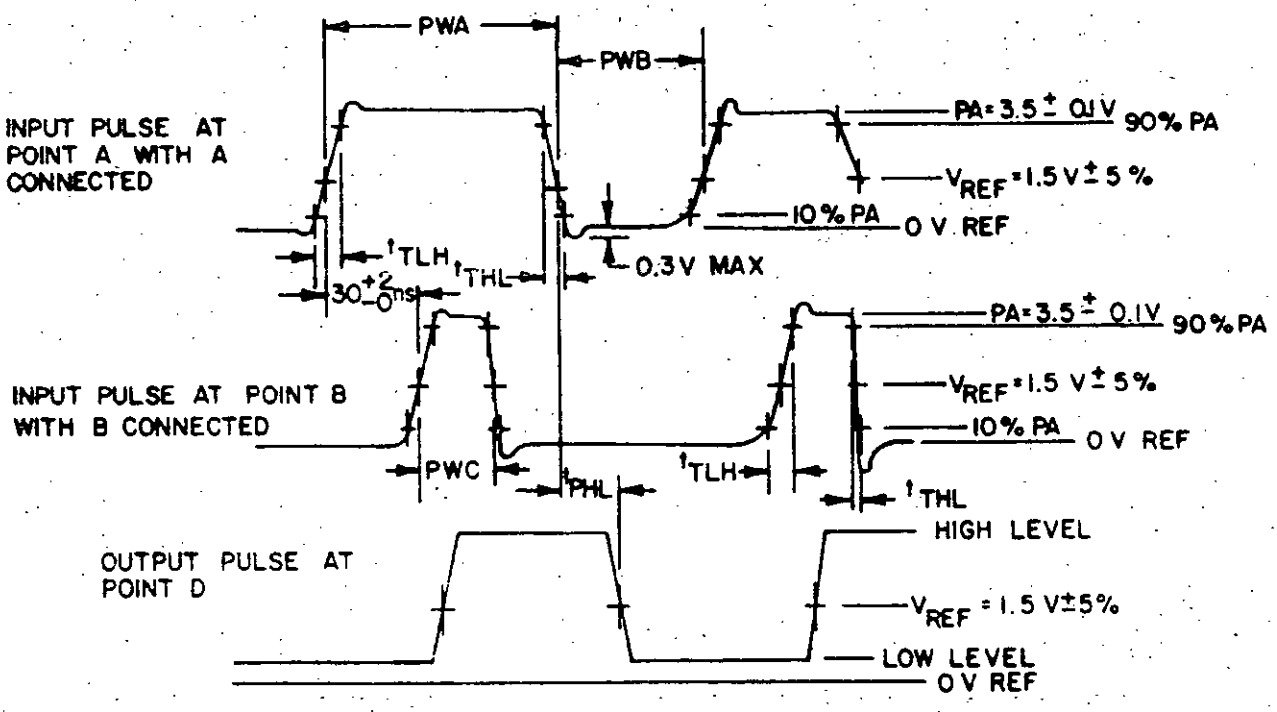
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. Load output pins 8, 12, 13, 14, 15, 21, 22, 25, 26, 27, 35, 36, 37, 38, and 39 with a load as specified on figure 10.

* FIGURE 18. Propagation delay time (t_{PHL}) test circuit for table XIV.

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NOTES:
1. Input pulse parameters:
   PWA = 80 ns ±10%.
   PWB = PWC = 30 +3, -0 ns.
   $t_{TLH} = t_{THL} = 8 \pm 2$ ns.
2. This test verifies a minimum reset pulse width of 30 ns and a minimum reset recovery time of 30 ns.

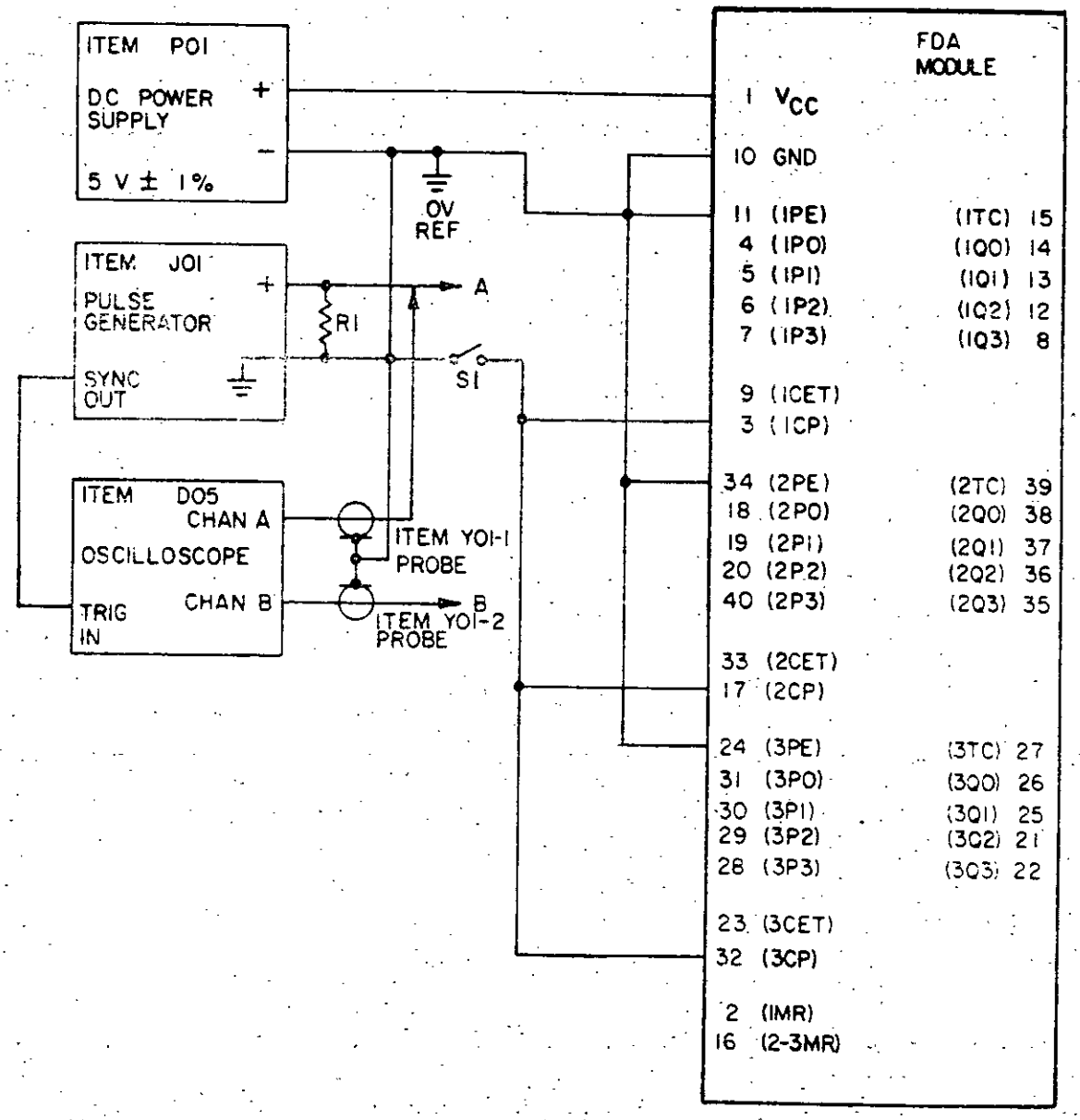
* FIGURE 19. Propagation delay time (t_{PHL}) test circuit waveforms for table XIV.

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TABLE XV. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X1	1CET to 1TC Propagation delay time (t_{PHL})	9	15	2	35	ns	20,21
X2	2CET to 2TC Propagation delay time (t_{PHL})	33	39		35		
X3	3CET to 3TC Propagation delay time (t_{PHL})	23	27		35		
Y1	1CET to 1TC Propagation delay time (t_{PLH})	9	15		40		
Y2	2CET to 2TC Propagation delay time (t_{PLH})	33	39		40		
Y3	3CET to 3TC Propagation delay time (t_{PLH})	23	27		40		

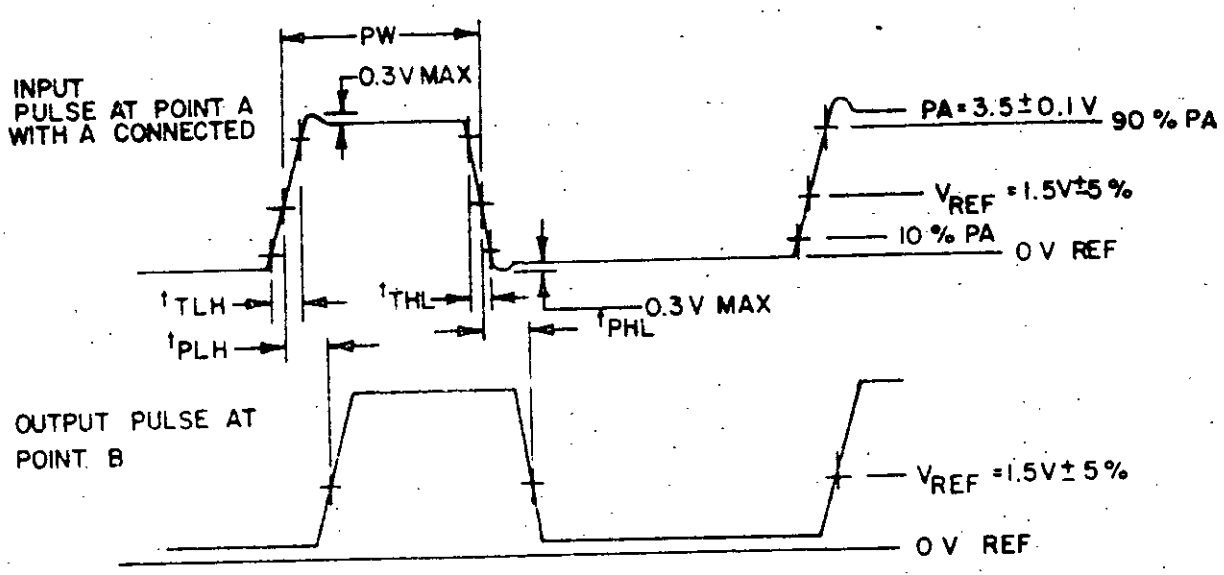
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. S1 = SPST switch.
3. Load output pins 15, 39, and 27 with a load as specified on figure 10.
4. Close S1 for at least 100 ns, open S1, then take measurement.

FIGURE 20. Propagation delay time (t_{PHL} and t_{PLH}) test circuit for table XV.

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NOTE: Input pulse parameters:
PW = 100 ns ±10%.
$t_{TLH} = t_{THL}$ = 8 ±2 ns.
PRR = 1 MHz ±10%.

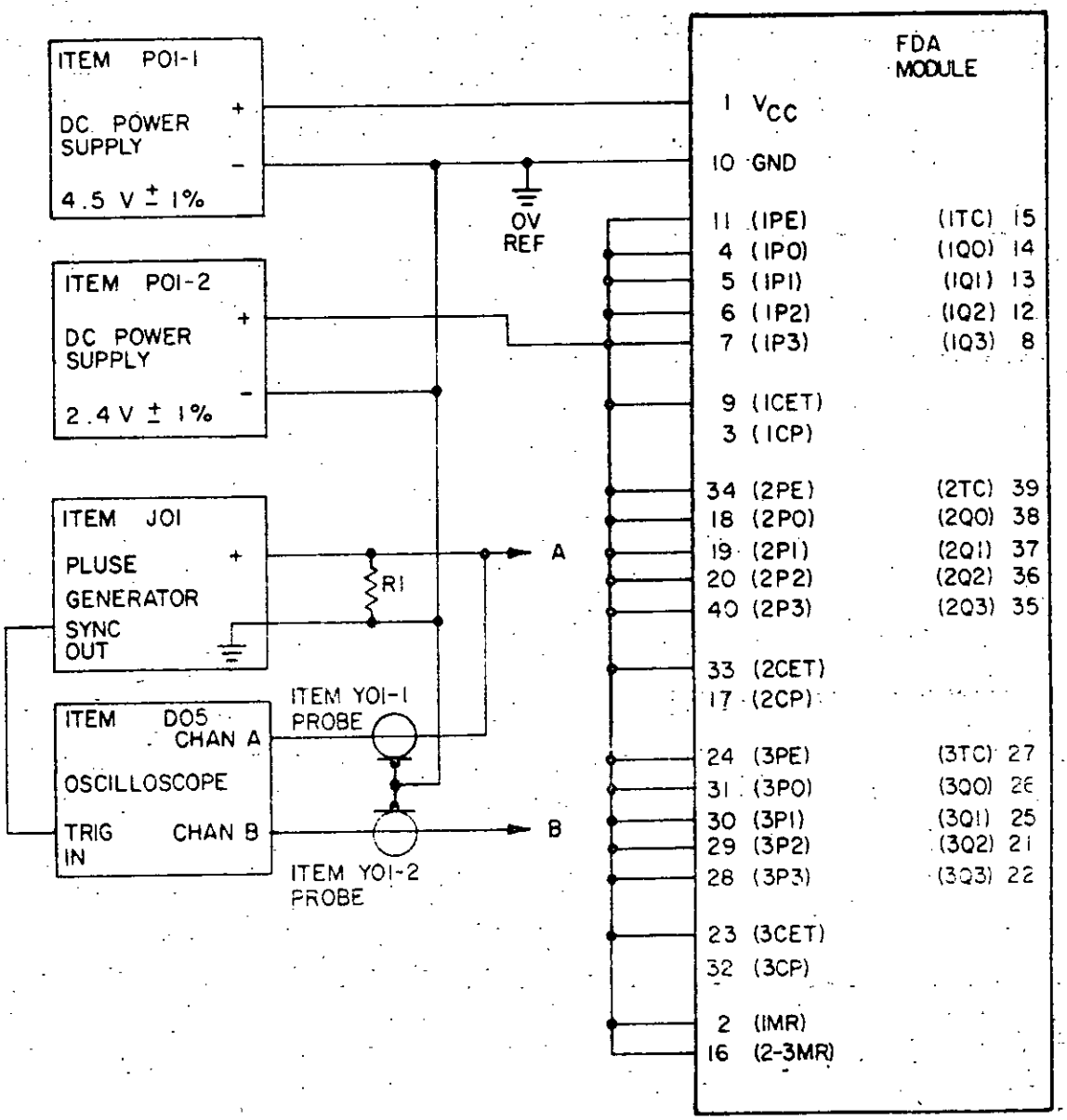
FIGURE 21. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for table XV.

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TABLE XVI. Maximum counting frequency test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Z1	1CP to 1Q3 Maximum counting frequency, PWA	3	8	460	500	ns	22,23
Z2	1CP to 1Q3 Maximum counting, frequency, PWB	3	8	↓	↓	↓	↓
Z3	2CP to 2Q3 Maximum counting, frequency, PWA	17	35				
Z4	2CP to 2Q3 Maximum counting, frequency, PWB	17	35				
Z5	3CP to 3Q3 Maximum counting, frequency, PWA	32	22				
Z6	3CP to 3Q3 Maximum counting, frequency, PWB	32	22				

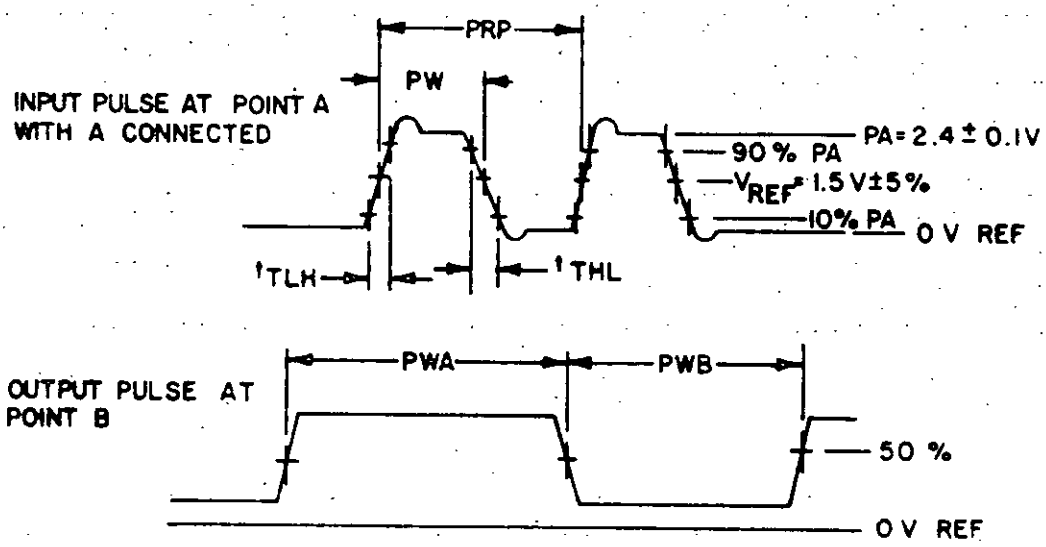
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Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Load output pins 8, 12, 13, 14, 15, 21, 22, 25, 26, 27, 35, 36, 37, 38, and 39 with a load as specified on figure 10.

* FIGURE 22. Maximum counting frequency test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   PW = 30 ±1 ns.
   PRP = 60 ±0.5 ns.
   $t_{TLH} = t_{THL} = 6 \pm 1$ ns.
2. The time scales shown for the waveforms at A and B are not the same.

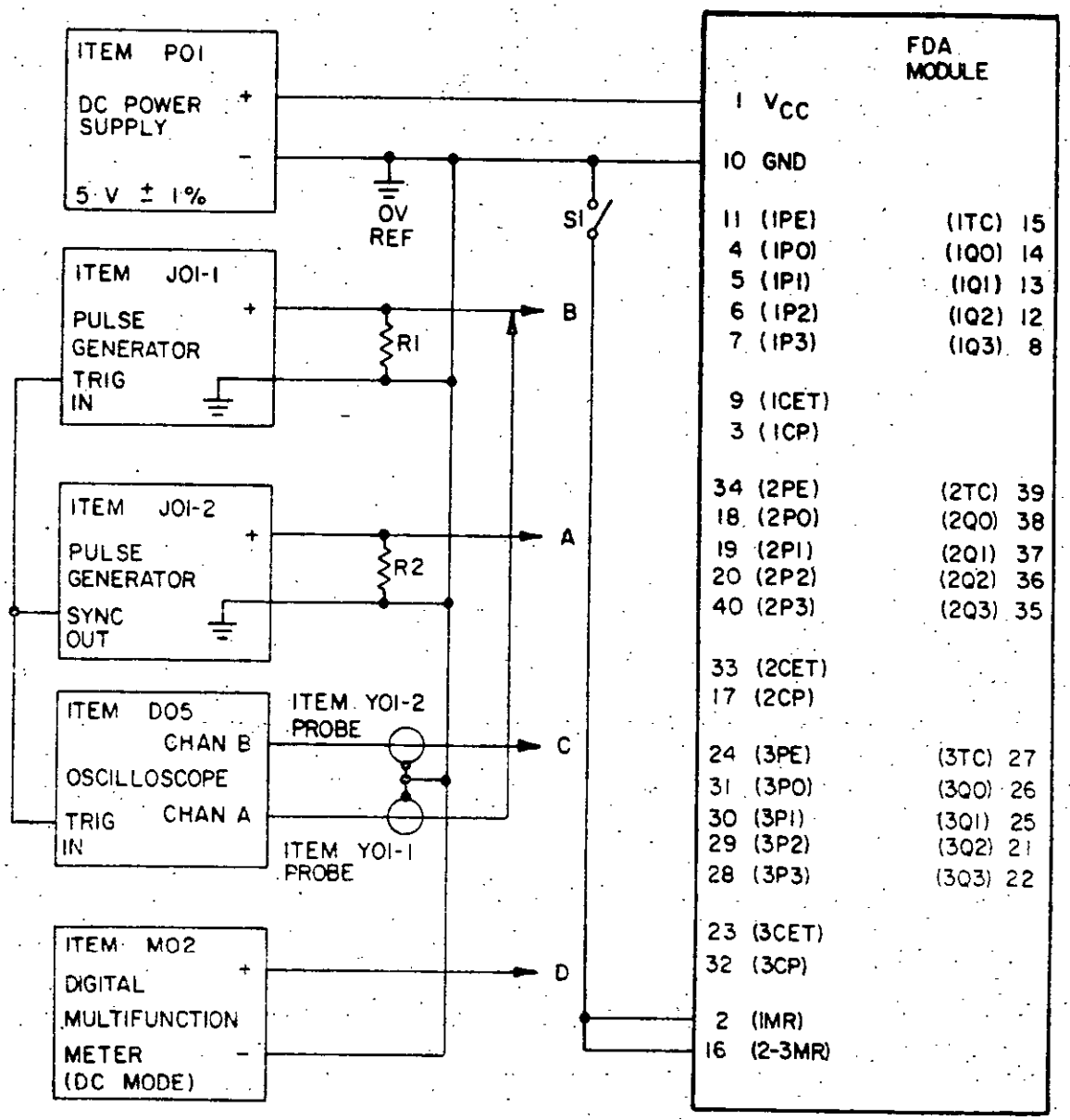
* FIGURE 23. Maximum counting frequency test circuit waveforms for table XVI.

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TABLE XVII. Setup and hold time (CET) check test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	t_U ns	Min	Max		
AA1	103 Setup time (CET) check	9	3	8	Open	25±1	15.6	16.4	µs	24,25
AA2	203 Setup time (CET) check	33	17	35	Open	25±1	15.6	16.4	µs	
AA3	303 Setup time (CET) check	23	32	22	Open	25±1	15.6	16.4	µs	
BB1	100 Hold time (CET) check	9	3	Open	14	0±1	0	400	mV	
BB2	200 Hold time (CET) check	33	17	Open	38	0±1	0	400	mV	
BB3	300 Hold time (CET) check	23	32	Open	26	0±1	0	400	mV	

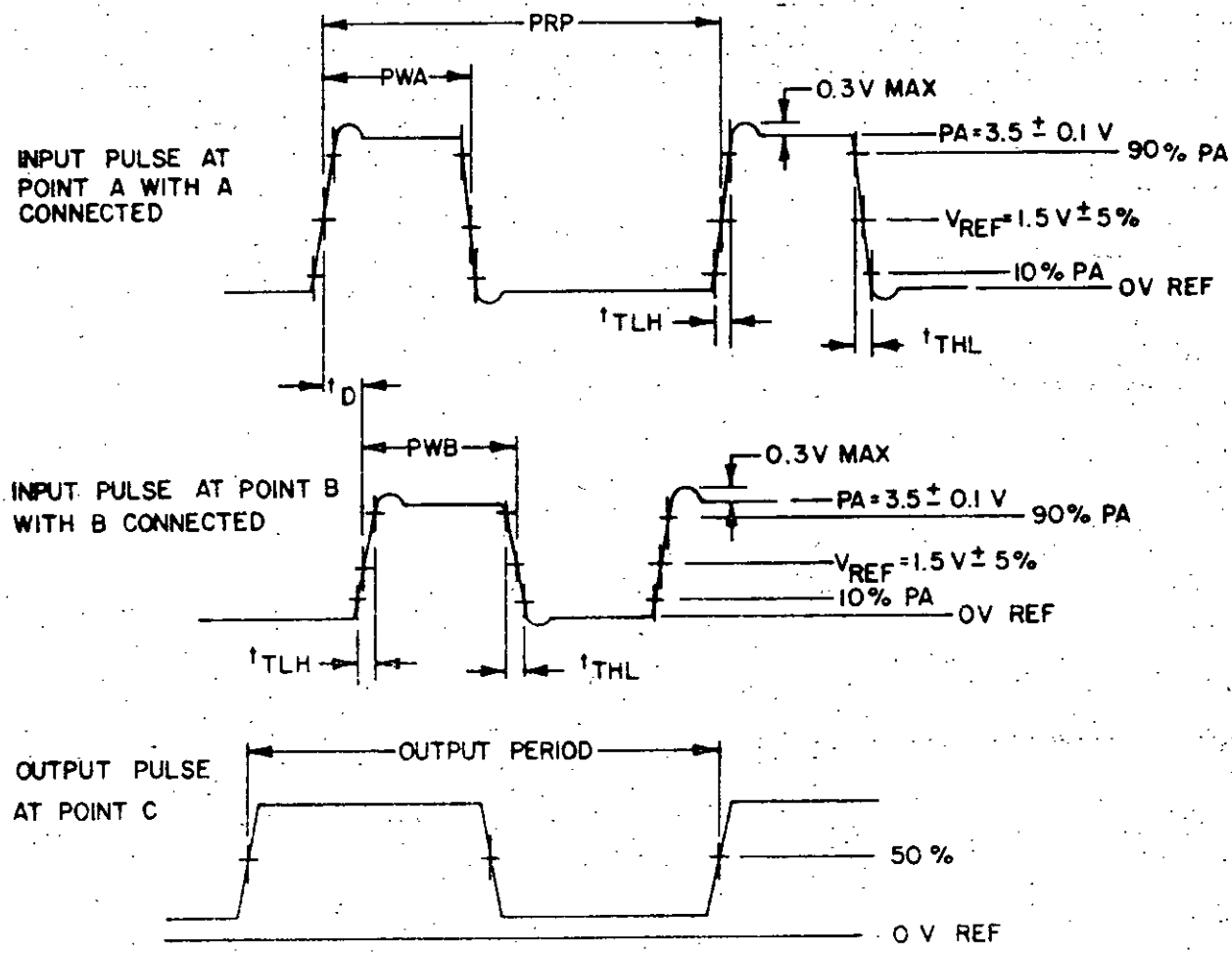
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. S1 = SPST switch.
3. After all circuit connections are made, close S1 for at least 100 ns, then open S1 prior to taking measurements.

* FIGURE 24. Setup and hold time (CET) check test circuit for table XVII.

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NOTES:
1. Input pulse parameters:
   PWA = 150 ns ±10%.
   PWB = 250 ns ±10%.
   PRP = 1 µs ±2%
   $t_{TLH} = t_{THL} = 8 \pm 2$ ns.
2. For setup times the output period shall be as specified in table XVII.

* FIGURE 25. Setup and hold time (CET) check test circuit waveforms for table XVII.

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TABLE XVIII. Setup and hold time (P and PE) check test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	t_D ns	Min	Max		
CC1	1Q0 Setup time (P) check	3	4	5,6,7, 11,9	14	40±1	0.8	1.2	µs	26,28
CC2	2Q0	17	18	19,20, 34,40, 33	38					
CC3	3Q0	32	31	24,28, 29,30, 23	26					
CC4	1Q1	3	5	4,6,7, 11,9	13					
CC5	1Q2	3	6	4,5,7, 11,9	12					
CC6	1Q3	3	7	4,5,6, 11,9	8					
CC7	2Q1	17	19	18,20, 34,40, 33	37					
CC8	2Q2	17	20	18,19, 34,40, 33	36					
CC9	2Q3	17	40	18,19, 20,34, 33	35					
CC10	3Q1	32	30	24,28, 29,31, 23	25					
CC11	3Q2	32	29	24,28, 30,31, 23	21					

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TABLE XVIII. Setup and hold time (P and PE) check test - Continued.

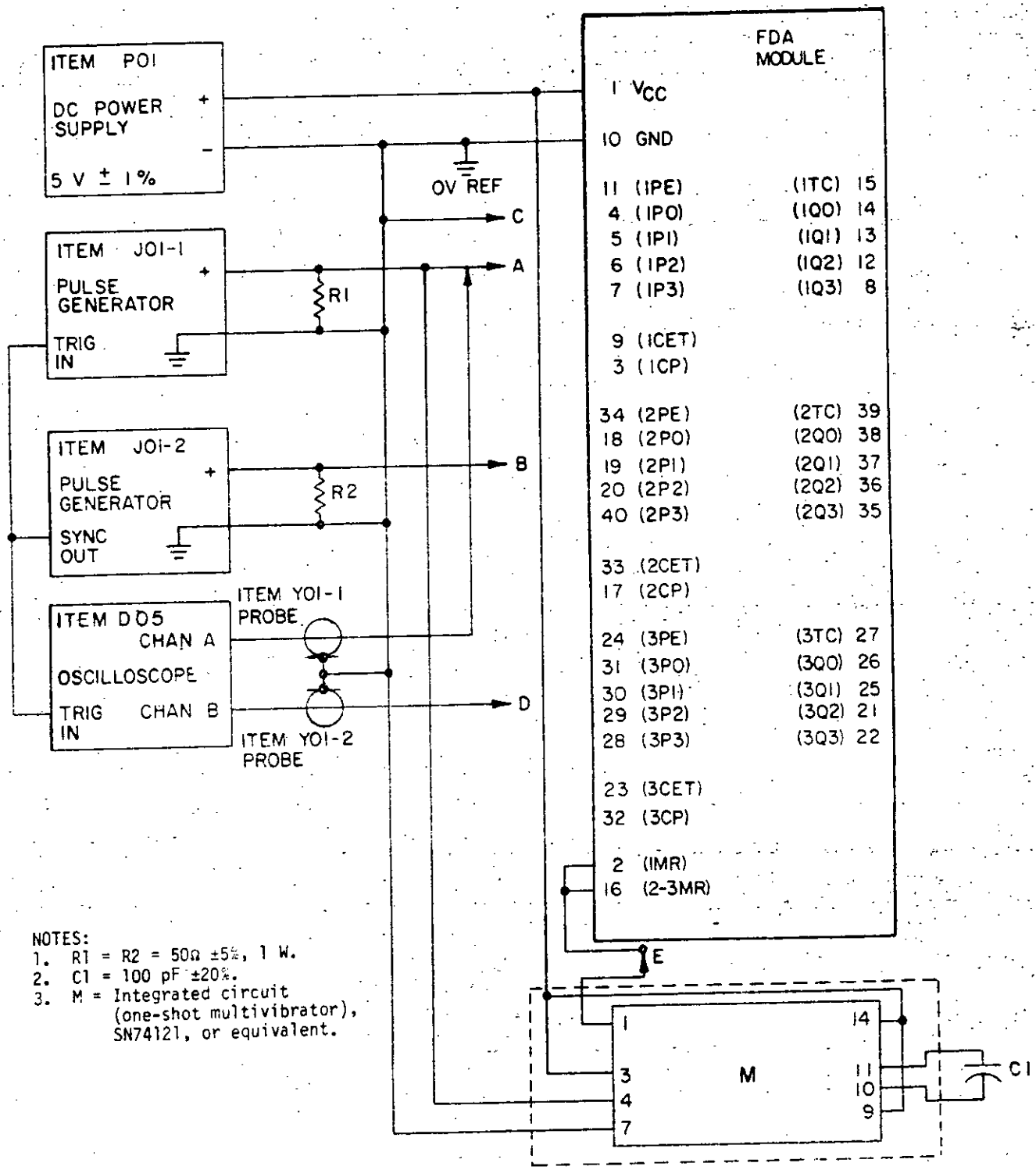
Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	t_D ns	Min	Max		
CC12	3Q3	Setup time (P) check	32	28	24,29, 30,31, 23	22	40±1	0.8	1.2	µs	26,28
DD1	1Q0	Hold time (P) check	3	4	14	11	5±1	0	400	mV	27,28
DD2	2Q0	↓	17	18	38	34	↓	↓	↓	↓	↓
DD3	3Q0		32	31	26	24					
DD4	1Q1		3	5	13	11					
DD5	1Q2		3	6	12	11					
DD6	1Q3		3	7	8	11					
DD7	2Q1		17	19	37	34					
DD8	2Q2		17	20	36	34					
DD9	2Q3		17	40	35	34					
DD10	3Q1		32	30	25	24					
DD11	3Q2		32	29	21	24					
DD12	3Q3		32	28	22	24					
EE1	1TC	Setup time (PE) check	3	11	Open	15	45±1	0.8	1.2	µs	26,29
EE2	2TC	Setup time (PE) check	17	34	Open	39	45±1	0.8	1.2	µs	26,29
EE3	3TC	Setup time (PE) check	32	24	Open	27	45±1	0.8	1.2	µs	26,29
FF1	1Q0	Hold time (PE) check	3	11	14	Open	7±1	0	400	mV	27,29
FF2	2Q0	Hold time (PE) check	17	34	38	Open	7±1	0	400	mV	27,29

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TABLE XVIII. Setup and hold time (P and PE) check test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	t_D ns	Min	Max		
FF3	3Q0	Hold time (PE) check	32	24	26	Open	7±1	0	400	mV	27,29
FF4	1Q1		3	11	13						
FF5	1Q2		3	11	12						
FF6	1Q3		3	11	8						
FF7	2Q1		17	34	37						
FF8	2Q2		17	34	36						
FF9	2Q3		17	34	35						
FF10	3Q1		32	24	25						
FF11	3Q2		32	24	21						
FF12	3Q3		32	24	22						

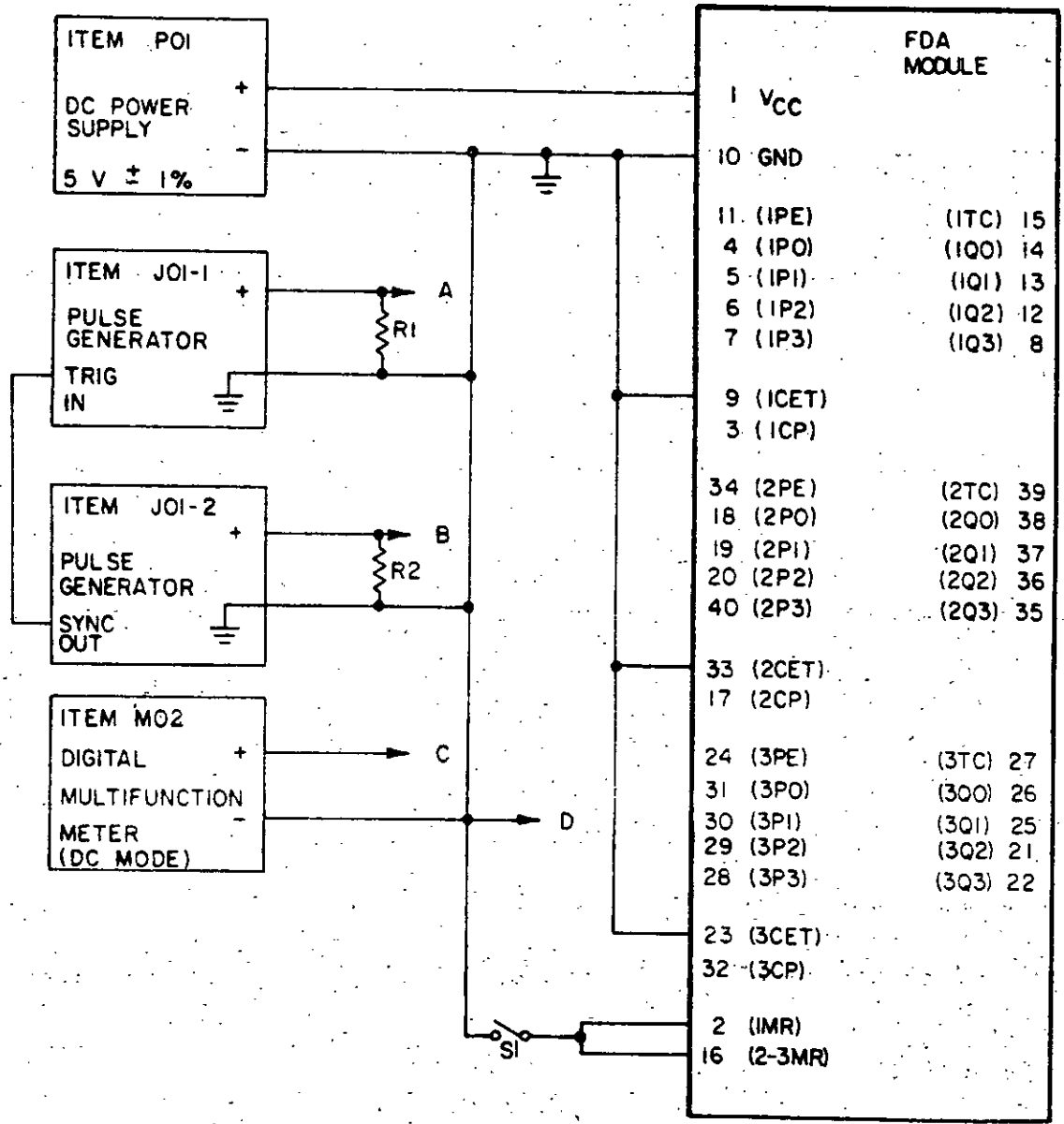
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. C1 = 100 pF ±20%.
3. M = Integrated circuit (one-shot multivibrator), SN74121, or equivalent.

* FIGURE 26. Setup time (P and PE) check test circuit for table XVIII.

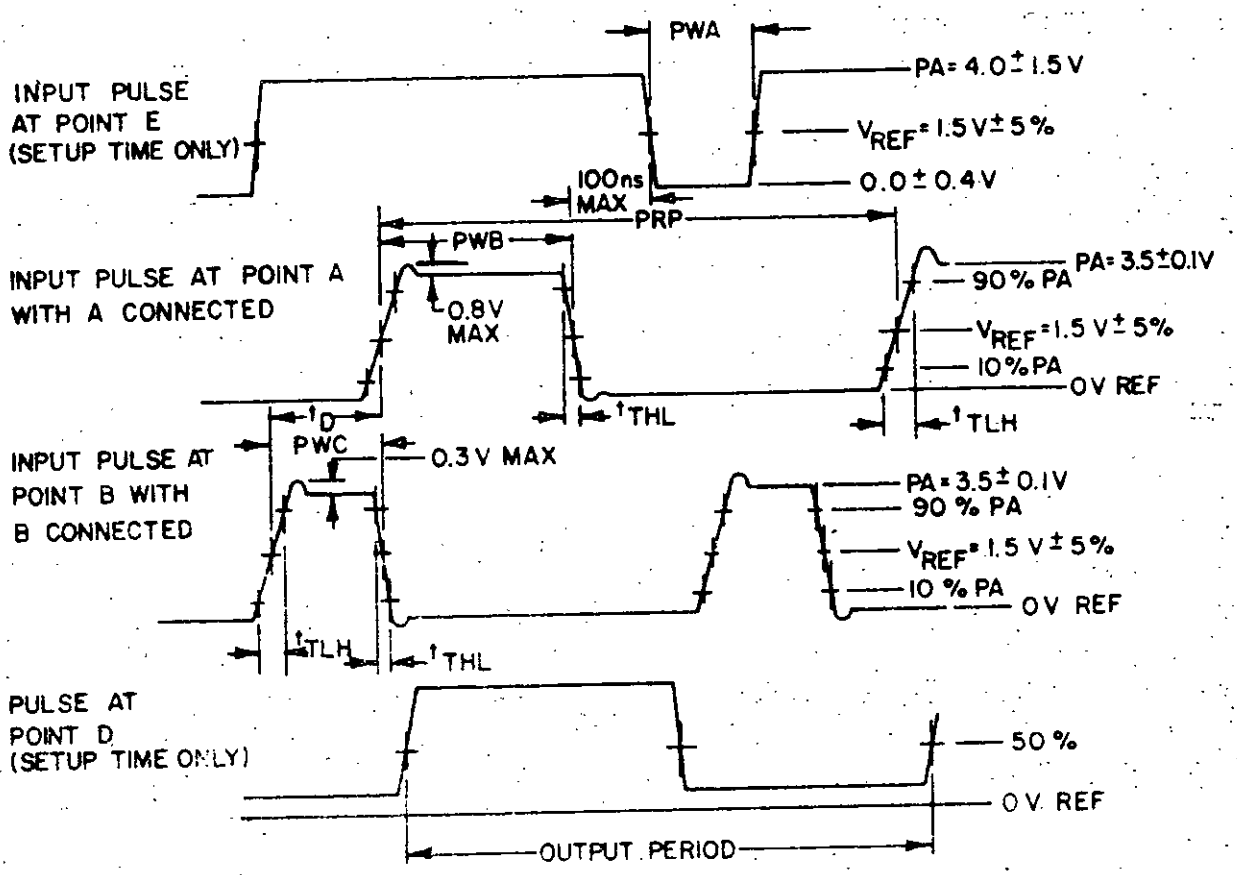
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. S1 = SPST switch.
3. Make all circuit connections for each test condition then: close S1 for at least 100 ns; open S1; take measurement.

* FIGURE 27. Hold time (P and PE) check test circuit for table XVIII.

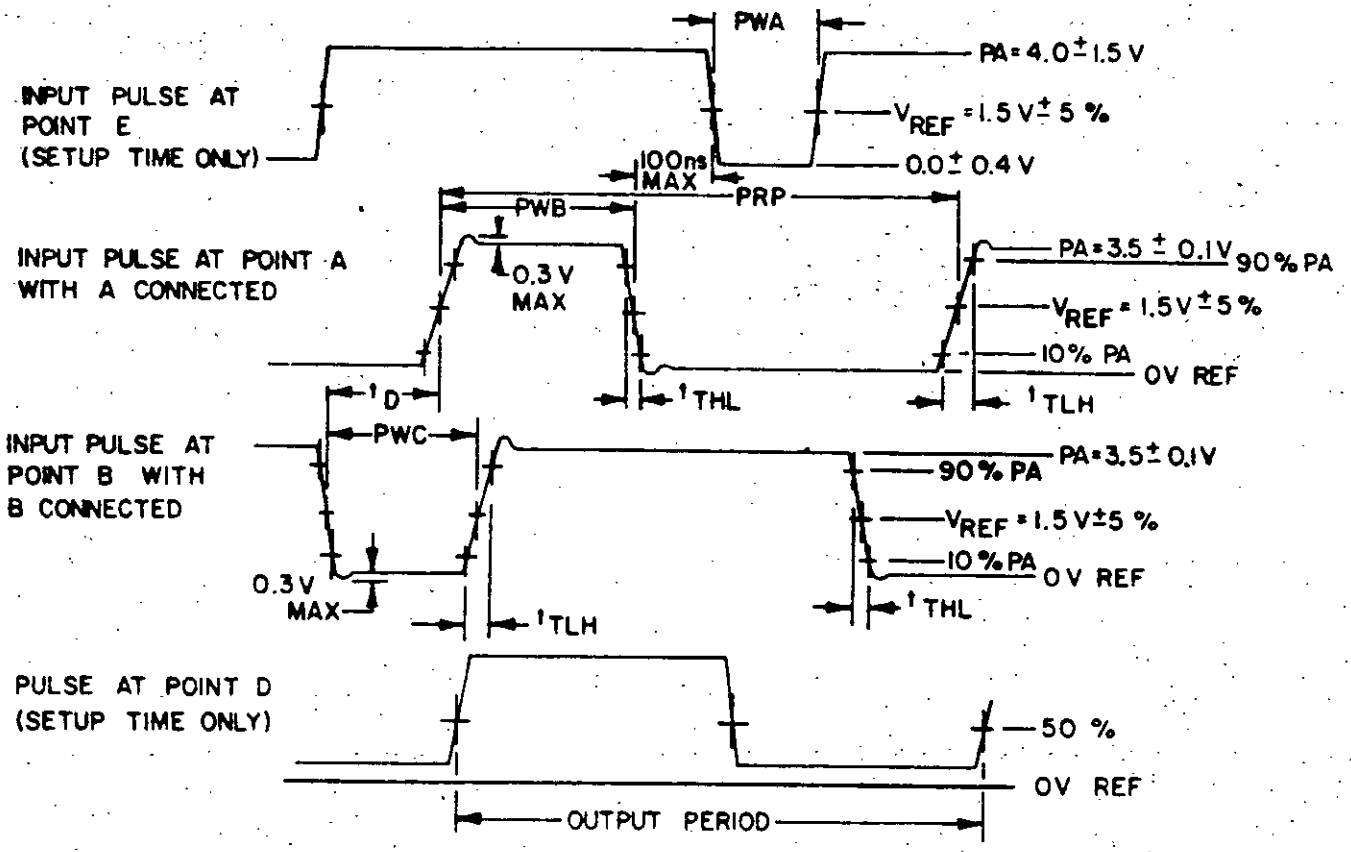
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NOTES:
1. Input pulse parameters:
   PWA = 100 ns ±50%.
   PWB = 200 ns ±10%.
   PWC = 100 ns ±10%.
   PRP = 1 µs ±10%.
   $t_{TLH} = t_{THL} = 8 \pm 2$ ns.
2. For setup times the output period shall be as specified in table XVIII.

* FIGURE 28. Setup and hold time (P) check test circuit waveforms for table XVIII.

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NOTES:
1. Input pulse parameters:
   PWA = 100 ns ±50%.
   PWB = 200 ns ±10%.
   PWC = 100 ns ±10%.
   PRP = 1 µs ±10%.
   $t_{TLH} = t_{THL} = 8 \pm 2$ ns.
2. For setup times the output period shall be as specified in table XVIII.

* FIGURE 29. Setup and hold time (PE) check test circuit waveforms for table XVIII.

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TABLE XIX. Output crosstalk (high-level and low-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	E to Pin	Min	Max		
GG1	1Q0 Output crosstalk, high level	4	Open	5,6,7,18,19,20,28,29, 30,31,40	14	0	400	mV	30,31
GG2	1Q1	5		4,6,7,18,19,20,28,29, 30,31,40	13				
GG3	1Q2	6		4,5,7,18,19,20,28,29, 30,31,40	12				
GG4	1Q3	7		4,5,6,18,19,20,28,29, 30,31,40	8				
GG5	1TC	9		18,19,20,28,29,30,31, 40	15				
GG6	2Q0	18		4,5,6,7,19,20,28,29, 30,31,40	38				
GG7	2Q1	19		4,5,6,7,18,20,28,29, 30,31,40	37				
GG8	2Q2	20		4,5,6,7,18,19,28,29, 30,31,40	36				
GG9	2Q3	40		4,5,6,7,18,19,20,28, 29,30,31	35				
GG10	2TC	33		4,5,6,7,28,29,30,31	39				
GG11	3Q0	31		4,5,6,7,18,19,20,28 29,30,40	26				
GG12	3Q1	30		4,5,6,7,18,19,20,28 29,31,40	25				

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TABLE XIX. Output crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	E to Pin	Min	Max		
GG13	3Q2 Output crosstalk, high level	29	Open	4,5,6,7,18,19,20,28 30,31,40	21	0	400	mV	30,31
GG14	3Q3 Output crosstalk, high level	28	Open	4,5,6,7,18,19,20,29, 30,31,40	22	0	400	mV	
GG15	3TC Output crosstalk, high level	23	Open	4,5,6,7,18,19,20,40	27	0	400	mV	
HH1	1Q0 Output crosstalk, low level	Open	4	5,6,7,18,19,20,28,29, 30,31,40	14	0.0	1.2	V	
HH2	1Q1		5	4,6,7,18,19,20,28,29, 30,31,40	13				
HH3	1Q2		6	4,5,7,18,19,20,28,29, 30,31,40	12				
HH4	1Q3		7	4,5,6,18,19,20,28,29, 30,31,40	8				
HH5	1TC		9	4,5,6,7,18,19,20,28, 29,30,31,40	15				
HH6	2Q0		18	4,5,6,7,19,20,28,29, 30,31,40	38				
HH7	2Q1		19	4,5,6,7,18,20,28,29, 30,31,40	37				
HH8	2Q2		20	4,5,6,7,18,19,20,28, 29,30,31,40	36				
HH9	2Q3		40	4,5,6,7,18,19,20,28, 29,30,31	35				

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TABLE XIX. Output crosstalk (high-level and low-level) test - Continued.

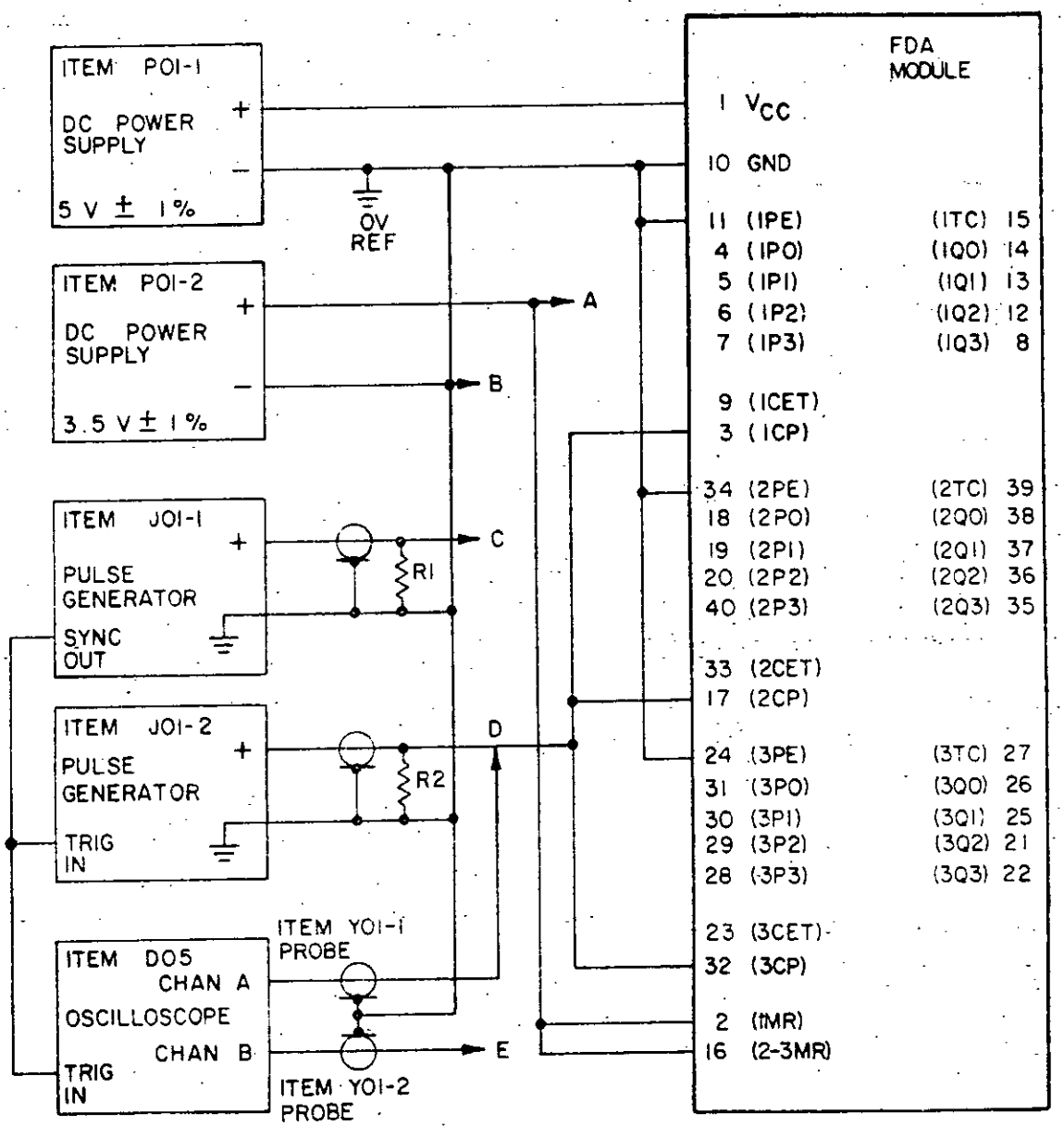
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	E to Pin	Min	Max		
HH10	2TC Output crosstalk, low level	Open	33	4,5,6,7,18,19,20,28, 29,30,31,40	39	0.0	1.2	V	30,31
HH11	3Q0		31	4,5,6,7,18,19,20,28, 29,30,40	26				
HH12	3Q1		30	4,5,6,7,18,19,20,28, 29,31,40	25				
HH13	3Q2		29	4,5,6,7,18,19,20,28, 30,31,40	21				
HH14	3Q3		28	4,5,6,7,18,19,20,29, 30,31,40	22				
HH15	3TC		23	4,5,6,7,18,19,20,28, 29,30,31,40	27				
HH16	1Q0		4	5,6,7,18,19,20,28, 29,30,31,40	14	0	400	mV	30,32
HH17	1Q1		5	4,6,7,18,19,20,28, 29,30,31,40	13				
HH18	1Q2		6	4,5,7,18,19,20,28, 29,30,31,40	12				
HH19	1Q3		7	4,5,6,18,19,20,28, 29,30,31,40	8				
HH20	1TC		9	4,5,6,7,18,19,20,28, 29,30,31,40	15				
HH21	2Q0		18	4,5,6,7,19,20,28, 29,30,31,40	38				

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TABLE XIX. Output crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	E to Pin	Min	Max		
HH22	2Q1 Output crosstalk, low level	Open	19	4,5,6,7,18,20,28,29, 30,31,40	37	0	400	mV	30,32
HH23	2Q2		20	4,5,6,7,18,19,28, 29,30,31,40	36				
HH24	2Q3		40	4,5,6,7,18,19,20,28, 29,30,31	35				
HH25	2TC		33	4,5,6,7,18,19,20,28, 29,30,31,40	39				
HH26	3Q0		31	4,5,6,7,18,19,20,28, 29,30,40	26				
HH27	3Q1		30	4,5,6,7,18,19,20,28, 29,31,40	25				
HH28	3Q2		29	4,5,6,7,18,19,20,28, 30,31,40	21				
HH29	3Q3		28	4,5,6,7,18,19,20,29, 30,31,40	22				
HH30	3TC		23	4,5,6,7,18,19,20,28, 29,30,31,40	27				

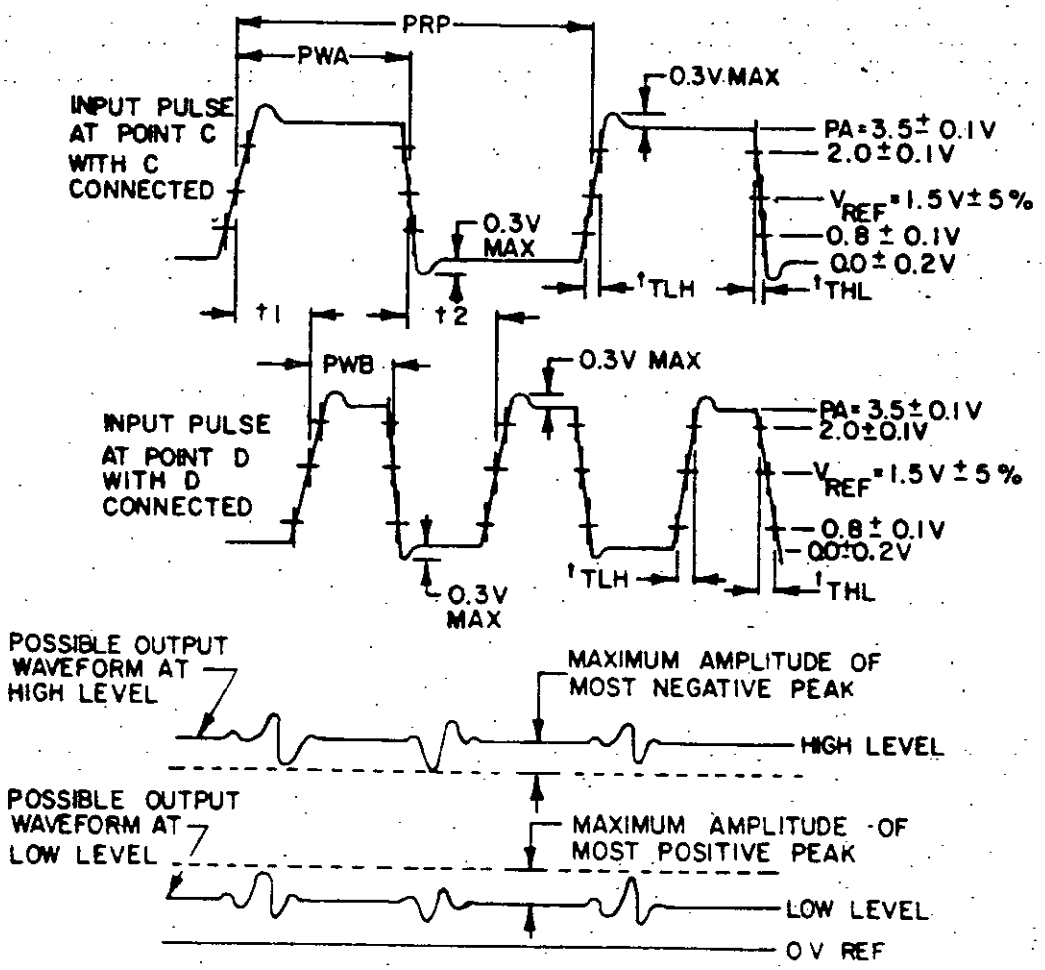
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. Load each output with load as specified on figure 10.

* FIGURE 30. Output crosstalk (high-level and low-level) test circuit for table XIX.

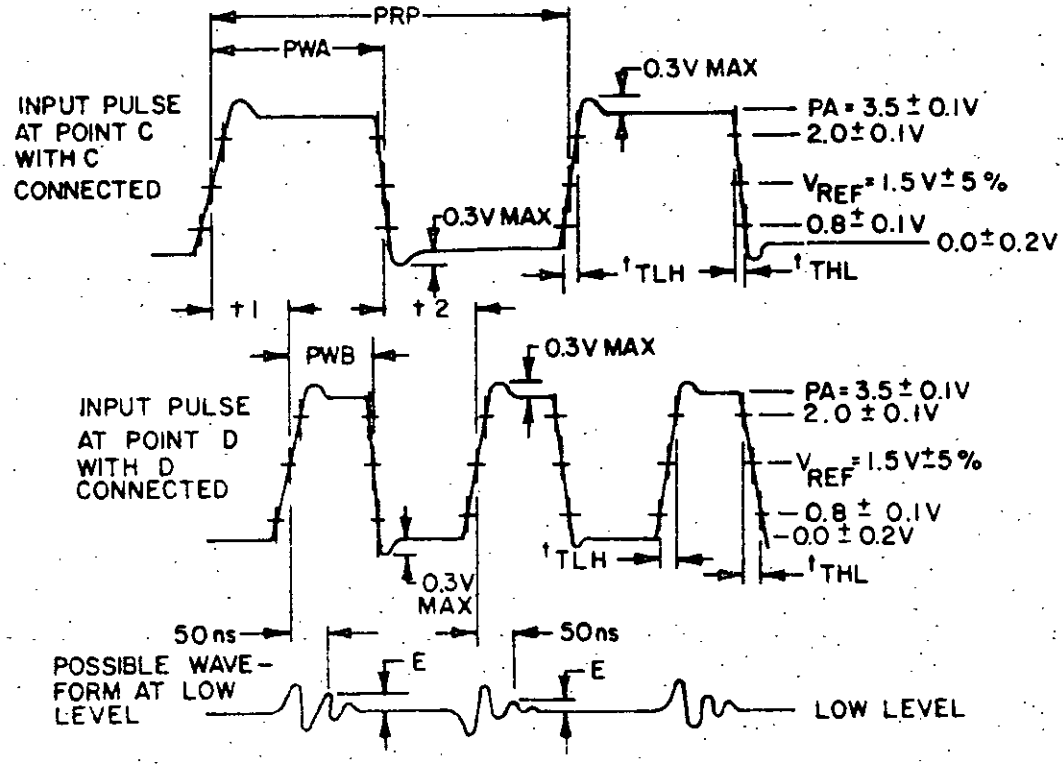
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NOTE: Input pulse parameters:
PWA = 200 ±20 ns.
PRP = 400 ±40 ns.
PWB = 100 ±10 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
t1 = t2 = 100 ±10 ns.

* FIGURE 31. Output crosstalk (high-level and low-level) test circuit waveforms for table XIX.

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NOTES:
1. Input pulse parameters:
   PWA = 200 ±20 ns.
   PRP = 400 ±40 ns.
   PWB = 100 ±10 ns.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
2. E is measured from the low level to the most positive peak which occurs at least 50 ns after the positive going edge of the clock pulse.

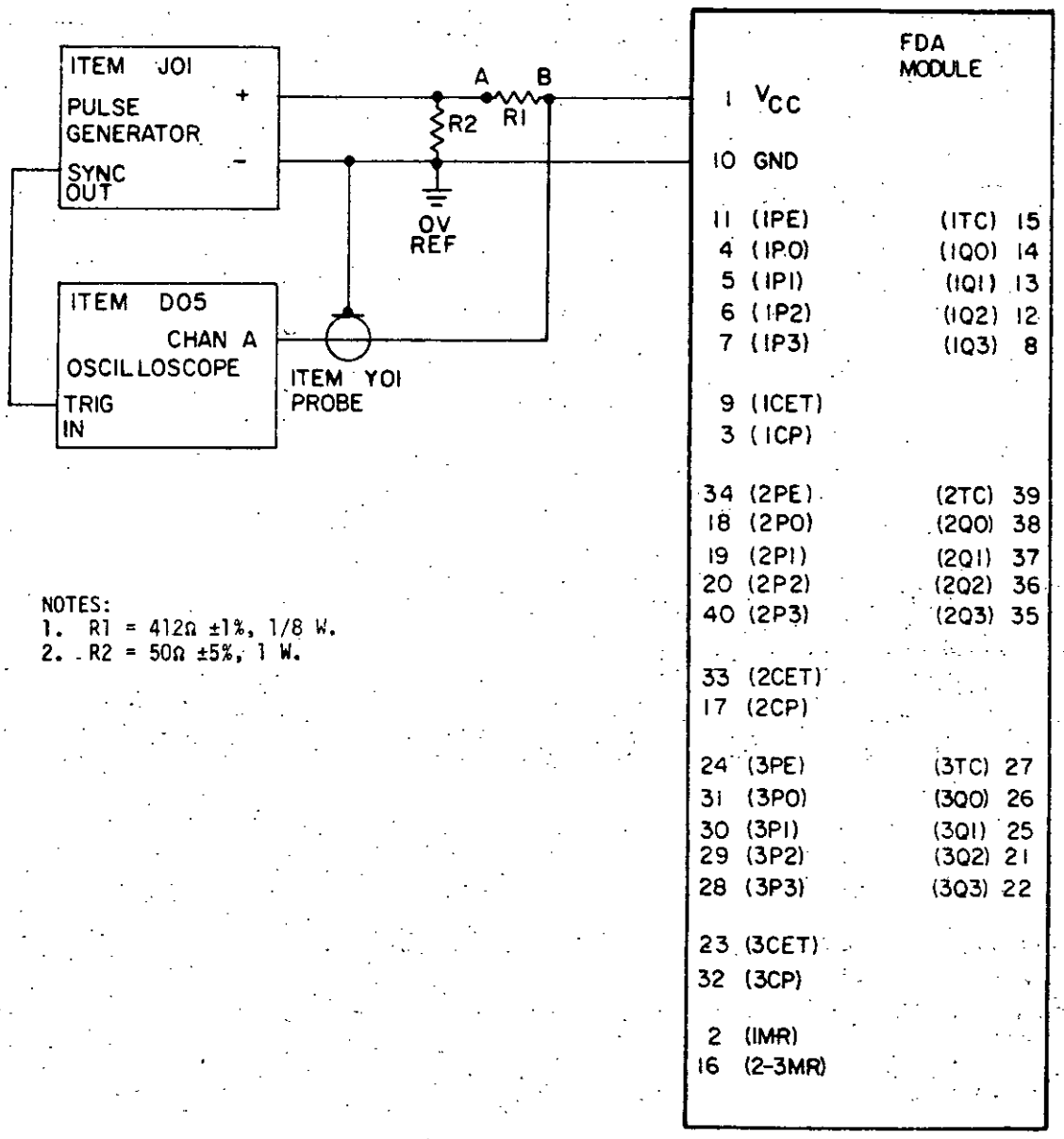
* FIGURE 32. Output crosstalk (low-level) test circuit waveforms for table XIX.

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TABLE XX. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
JJ	Power supply transient rise time	2.5	7.5	µs	33,34

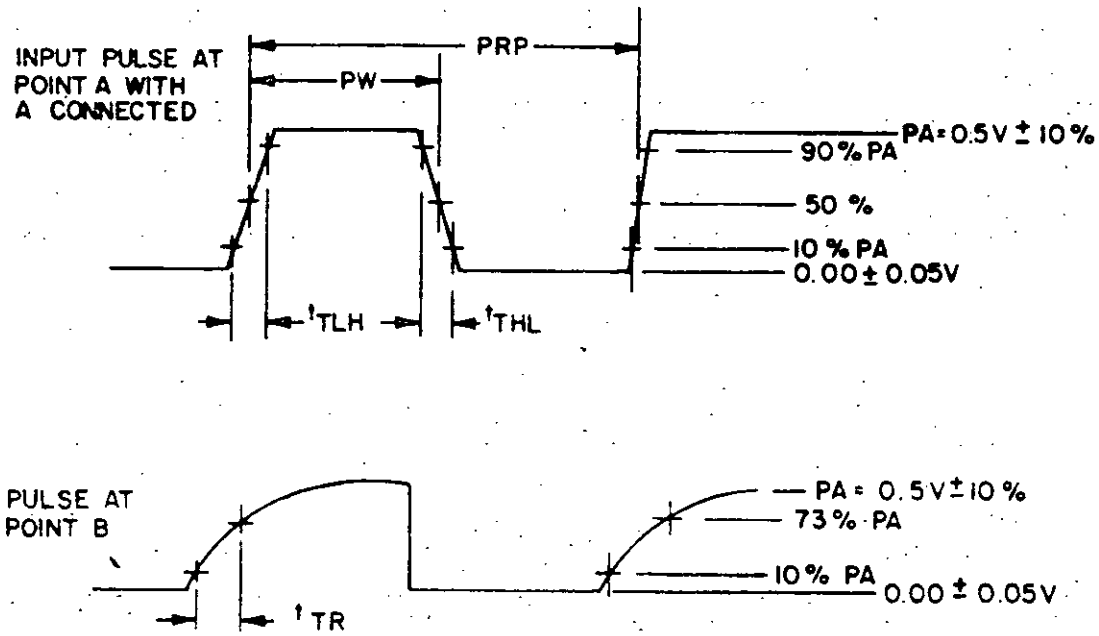
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NOTES:
1. R1 = 412Ω ±1%, 1/8 W.
2. R2 = 50Ω ±5%, 1 W.

FIGURE 33. Power supply transient rise time test circuit for table XX.

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NOTES:
1. Input pulse parameters:
   PW = 30 µs ±10%.
   $t_{TLH} = t_{THL}$ = 50 ns ±10%.
   PRP = 60 µs ±10%.
2. Measure t_{TR} at point B.

* FIGURE 34. Power supply transient rise time test circuit waveforms for table XX.

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